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# Statistical Modelling

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# Notice

- Slides prepared by Moayad Alnammi – [malnammi.github.io](https://malnammi.github.io)
- Some figures adapted from CS188: Introduction to Artificial Intelligence, UC Berkeley (Dan Klein, Pieter Abbeel) — [ai.berkeley.edu](https://ai.berkeley.edu)
- Some figures are obtained from the [AIMA Book](#)
- Some content adapted from Dr. ElSayed El-Alfy
- Some content adapted from Dr. Richard Lathrop (UC Irvine)

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# Outline

- Review of probability concepts
- Review of distributions: discrete and continuous, univariate vs multivariate
- How to sample from a distribution
- Estimating parameters of a distribution using data (maximum likelihood estimation)
- Statistical modelling of a problem with many variables:
  - Full-joint distribution, exact inference, parameter estimation
  - Reducing complexity by exploiting assumptions: independence, conditional independence
  - Bayesian networks, exact inference, approximate inference, parameter estimation

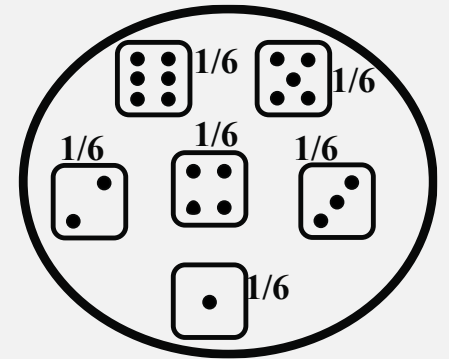
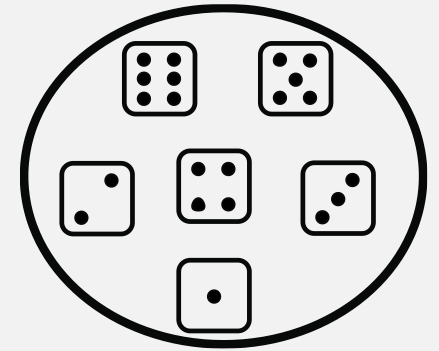
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# Probability Review

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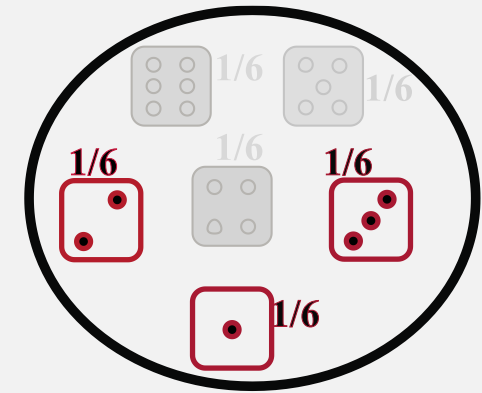
# Probability Basics [1]

- **Sample space**  $\Omega$  is the set of all possible outcomes
  - E.g., 6 possible rolls of a die,  $\Omega = \{1, 2, 3, 4, 5, 6\}$
- A **sample/outcome** is an element  $\omega \in \Omega$ .
- A **probability model** assigns a number  $P(\omega)$  to each outcome  $\omega$ 
  - E.g.,  $P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = 1/6$ .
- Probability model must satisfy
  - $0 \leq P(\omega) \leq 1$
  - $\sum_{\omega \in \Omega} P(\omega) = 1$



# Probability Basics [2]

- An **event**  $A$  is any subset of  $\Omega$ .  $P(A) = \sum_{\omega \in A} P(\omega)$ 
  - e.g.,  $P(\text{die roll} < 4) = P(1) + P(2) + P(3) = 1/6 + 1/6 + 1/6 = 1/2$
- $P(\neg A) = 1 - P(A)$     -- **complement event**
- Special cases
  - $P(A) = 0$  iff  $A = \emptyset$     -- **impossible/empty event**
  - $P(A) = 1$  iff  $A = \Omega$     -- **certain event**
- $P(A \vee B) = P(A) + P(B) - P(A \wedge B)$
- If  $A$  and  $B$  are **disjoint** events (**mutually exclusive**; occurrence of one excludes the other, i.e. can not happen at the same time):
  - $P(A \wedge B) = 0$
  - $P(A \vee B) = P(A) + P(B)$



# Random Variables

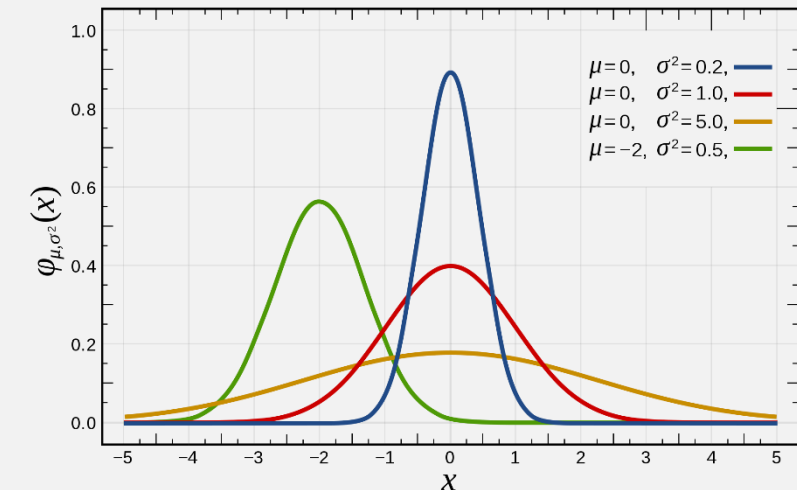
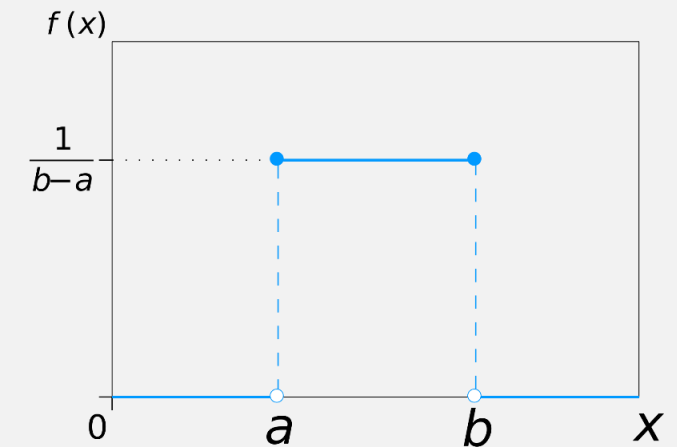
- **Random variable:** For ease of syntax, variables are used to denote aspects of the world for which we would like to model uncertainty (probabilities).
- Variables can be assigned values that represent the outcome of an event.
- Variables can be discrete (obtainable by counting), or continuous (obtainable by measuring). Distinction does not matter too much for now. For now, be aware of common discrete and continuous distributions.
- When we say that the variable  $X$  follows distribution  $D$ . We denote this as  $X \sim D$
- Discrete Example: Coin flip
  - Variable  $R$ , the result of the coin flip
  - $R \in \{heads, tails\}$
  - $P(R = heads) = 0.2$
  - $P(R = tails) = 0.8$
- Continuous Example: Blood sugar
  - Variable  $B$ , blood sugar value
  - $B \in \mathbb{R}$
  - $P(70 < B < 100) = 0.8$
  - $P(B < 70) = 0.1$
  - $P(B > 100) = 0.1$

# Example Discrete Distribution – Categorical and Bernoulli

- A random variable  $X$  is said to be Categorically distributed, if it can take on a finite set of  $n$  outcomes  $\{x_1, \dots, x_k\}$ ; typically outcomes are just mapped to  $\{0, 1, \dots, k - 1\}$  or  $\{1, 2, \dots, k\}$ .
- We denote it as  $X \sim \text{Cat}([p_1, p_2, \dots, p_k])$  where  $p_i \in [0,1]$  is the probability of outcome  $x_i$  and  $\sum_i p_i = 1$
- When all  $p_i = \frac{1}{n}$  it is said to be *uniformly* distributed.
- When  $n = 2$ , this is just the *Bernoulli*( $p$ ) distribution with  $p \in [0,1]$  and two outcomes  $\{0, 1\}$ .
- A process with two outcomes example: flipping a two-sided fair coin
  - $x \in \{0, 1\}$  and  $x \sim \text{Bernoulli}(p = 0.5)$
  - Common notation is that  $P(x = 1) = p$  and  $P(x = 0) = 1 - p$
- A process with 4 outcomes example: pulling a colored ball out of a bag (red, green, blue, yellow)
  - $x \in \{0, 1, 2, 3\}$  and  $x \sim \text{Cat}(p = [0.2, 0.3, 0.4, 0.1])$
- **Note:** distributions have parameters. For Cat, the  $p_i$  are the parameters. For Bernoulli, the  $p$  is the parameter.

# Example Continuous Distribution – Uniform and Normal

- Express distribution as a parameterized function of value. This function is called **probability density function**  $f(x)$ .
- For continuous variables, the  $P(v_1 \leq X \leq v_2) = \int_{v_1}^{v_2} f(x)dx$
- It must be the case that:  $f(x) \geq 0$  for all  $x$  and  $\int_{-\infty}^{+\infty} f(x)dx = 1$
- Common continuous distributions:
  - Uniform(a, b);  $f(x) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$
  - Normal( $\mu, \sigma$ );  $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$
- What are parameters of the uniform distribution? What about the normal?



Images from: [link1](#) and [link2](#)

# Probability Model and Full-Joint Distribution

- Consider an environment with Temperature and Weather.
- We can construct a **probability model** for this environment using two random variables.
- The **full-joint distribution** over all random variables in a **probability model** gives us all information required to compute any event.

Probability that it's hot AND sunny?  $P(T=\text{hot}, W=\text{sun})$

Probability that it's hot?  $P(T = \text{hot})$

Probability that it's hot OR not foggy?  $P(T=\text{hot OR } W \neq \text{fog})$   
 $P(A \vee B) = P(A) + P(B) - P(A \wedge B)$

*Joint distribution*  
 $P(T,W)$   
shorthand for  $P(T \text{ and } W)$

|         |        | Temperature |      |
|---------|--------|-------------|------|
|         |        | hot         | cold |
| Weather | sun    | 0.45        | 0.15 |
|         | rain   | 0.02        | 0.08 |
|         | fog    | 0.03        | 0.27 |
|         | meteor | 0.00        | 0.00 |

# Marginal Distribution

- **Marginal distribution** refers to the probabilities of a subset of variables.
- **Marginalization (summing out)**: is the process of summing out variables from a joint probability.

$P(X=x) = \sum_y P(X=x, Y=y)$  for model with two vars X, Y

$P(X=x) = \sum_y \sum_z P(X=x, Y=y, Z=z)$  with three vars X, Y, Z

$P(X=x, Y=y) = \sum_z P(X=x, Y=y, Z=z)$  with three vars X, Y, Z

- Consider the joint probability table for the temperature and weather model.

What is the probability that it's hot?

$$\begin{aligned} P(T = hot) &= \sum_w P(T = hot, W = w) \\ &= 0.45 + 0.02 + 0.03 + 0.00 = 0.5 \end{aligned}$$

$P(T, W)$

|         |        | Temperature |      |      |
|---------|--------|-------------|------|------|
|         |        | hot         | cold |      |
| Weather | sun    | 0.45        | 0.15 | 0.60 |
|         | rain   | 0.02        | 0.08 | 0.10 |
|         | fog    | 0.03        | 0.27 | 0.30 |
|         | meteor | 0.00        | 0.00 | 0.00 |
|         |        | 0.50        | 0.50 |      |

$P(T)$

$P(W)$

# Conditional Probability [1]

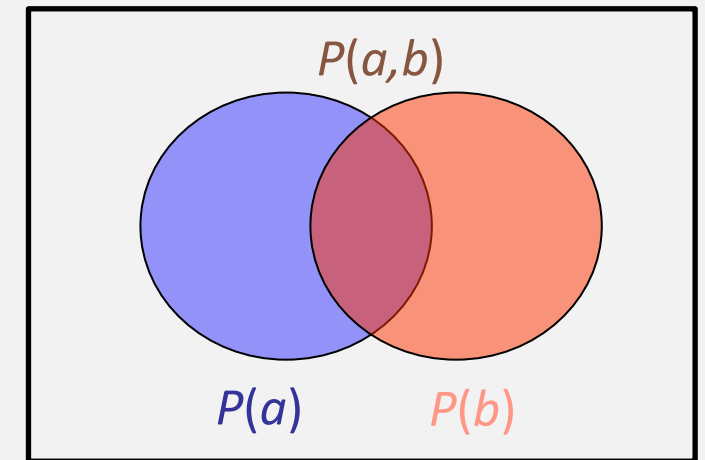
- $P(A|B)$  is the **conditional** or **posterior** probability of event A, **given** that event B has occurred (B is also referred to as **evidence**).
  - Reflects the updated probability that A is true, after knowing B  
e.g.  $P(\text{rain in Riyadh tomorrow} \mid \text{raining in Riyadh today})$
- Note that conditional probabilities obey the same rule as non-conditional probabilities.
  - $P(A = a \mid B = b) = 1 - P(A \neq a \mid B = b)$
  - $\sum_{a \in A} P(A = a \mid B = b) = 1$

- A simple relation between joint and conditional probabilities is given by:

$$P(A|B) = \frac{P(A, B)}{P(B)}$$

- Alternatively, for the joint probability (called **Product Rule**):

$$P(A, B) = P(A|B)P(B) = P(B|A)P(A)$$

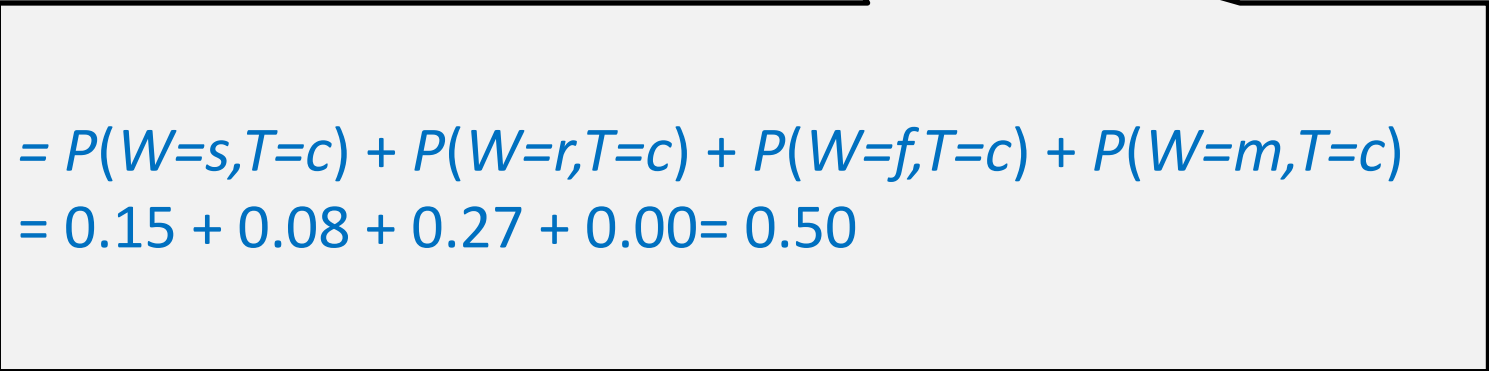


# Conditional Probability [2]

- Consider the joint probability table for the temperature and weather model.

What is the probability that it's sunny given that we know it's cold?

$$P(W = s \mid T = c) = \frac{P(W = s, T = c)}{P(T = c)} = \frac{0.15}{0.50} = 0.3$$


$$\begin{aligned} &= P(W=s, T=c) + P(W=r, T=c) + P(W=f, T=c) + P(W=m, T=c) \\ &= 0.15 + 0.08 + 0.27 + 0.00 = 0.50 \end{aligned}$$

$$P(A|B) = \frac{P(A, B)}{P(B)}$$

|         |        | Temperature |      |
|---------|--------|-------------|------|
|         |        | hot         | cold |
| Weather | sun    | 0.45        | 0.15 |
|         | rain   | 0.02        | 0.08 |
|         | fog    | 0.03        | 0.27 |
|         | meteor | 0.00        | 0.00 |

# Chain Rule

- A joint distribution can be written as a product of conditional distributions by **repeated** application of the product rule:  $P(A, B) = P(A|B)P(B)$

$$\begin{aligned} P(x_1, x_2, x_3) &= P(x_3|x_1, x_2)P(x_1, x_2) \\ &= P(x_3|x_1, x_2)P(x_2|x_1)P(x_1) \end{aligned}$$

- General form of chain rule:

$$P(x_1, x_2, \dots, x_n) = p(x_1) \prod_{i=2}^n P(x_i|x_1, \dots, x_{i-1})$$

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# Distributions Review

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# Random Variable Formalism Review

- Nice reference lookup [here](#).
- Here are some more formal terms because we will mention it later when we talk about sampling.
- An RV has a **probability mass function PMF** (discrete case); **probability density function PDF** (continuous case).
- See formal definition of PMF [here](#).
- See formal definition of PDF [here](#).
- Given a pmf/pdf  $f(x)$ , we get the **cumulative distribution function CDF**  $F(x)$ . See [here](#).

$$F(x) = \Pr(X \leq x)$$

- The cdf in particular is of interest, because we can use it to *sample* data from a given distribution. More on this later.

# Expectation and Variance

- The **expectation (expected value, mean, average)** of a random variable is its central tendency. It is the weighted average where the probabilities are the weights. Denoted as  $E[X]$  or  $\mu$  or  $\mu_X$
- **Variance** of a random variable is a measure of its spread or dispersion from its central tendency. Denoted as  $Var(X)$  or  $\sigma^2$ . Standard deviation is  $\sigma$ .

$$Var[X] = E[(X - \mu)^2] = E[X^2] - \mu^2$$

- For discrete RV:  $E[X] = \sum_x xp(x)$

$$Var[X] = \sum_x (x - \mu)^2 p(x)$$

- For continuous RV:  $E[X] = \int_x xf(x)dx$

$$Var[X] = \int_x (x - \mu)^2 f(x)dx$$

# Notes on Univariate vs Multivariate

Let  $X$  be a random variable with possible outcomes  $\mathcal{X}$ . Then,

- $X$  is called a two-valued random variable or random event, if  $\mathcal{X}$  has exactly two elements, e.g.  $\mathcal{X} = \{E, \overline{E}\}$  or  $\mathcal{X} = \{\text{true}, \text{false}\}$  or  $\mathcal{X} = \{1, 0\}$ ;
- $X$  is called a univariate random variable or random scalar, if  $\mathcal{X}$  is one-dimensional, i.e. (a subset of) the real numbers  $\mathbb{R}$ ;
- $X$  is called a multivariate random variable or random vector, if  $\mathcal{X}$  is multi-dimensional, e.g. (a subset of) the  $n$ -dimensional Euclidean space  $\mathbb{R}^n$ ;
- $X$  is called a matrix-valued random variable or random matrix, if  $\mathcal{X}$  is (a subset of) the set of  $n \times p$  real matrices  $\mathbb{R}^{n \times p}$ .
- Above taken from [link](#)
- The expectation formula is the same for all cases. The main difference is that  $E[X] = \mu$  can be vector/matrix.
- What about variance? In case of random vector, it is a matrix and typically called Covariance.

# Multivariate Expectation

Let  $X$  be an  $n \times 1$  random vector. Then, the expected value of  $X$  is an  $n \times 1$  vector whose entries correspond to the expected values of the entries of the random vector:

$$E(X) = E \left( \begin{bmatrix} X_1 \\ \vdots \\ X_n \end{bmatrix} \right) = \begin{bmatrix} E(X_1) \\ \vdots \\ E(X_n) \end{bmatrix}$$

Let  $X$  be an  $n \times p$  random matrix. Then, the expected value of  $X$  is an  $n \times p$  matrix whose entries correspond to the expected values of the entries of the random matrix:

$$E(X) = E \left( \begin{bmatrix} X_{11} & \dots & X_{1p} \\ \vdots & \ddots & \vdots \\ X_{n1} & \dots & X_{np} \end{bmatrix} \right) = \begin{bmatrix} E(X_{11}) & \dots & E(X_{1p}) \\ \vdots & \ddots & \vdots \\ E(X_{n1}) & \dots & E(X_{np}) \end{bmatrix}$$

- Above taken from [link](#) and [link](#).
- **Note:** each element can be discrete or continuous random variable.

# Multivariate Variance (Covariance)

The covariance of two random variables  $X$  and  $Y$  is defined as the expected value of the product of their deviations from their individual expected values:

$$\text{Cov}(X, Y) = \text{E} [(X - \text{E}[X])(Y - \text{E}[Y])]$$

Let  $X = [X_1, \dots, X_n]^T$  be a random vector. Then, the covariance matrix of  $X$  is defined as the  $n \times n$  matrix in which the entry  $(i, j)$  is the covariance of  $X_i$  and  $X_j$ :

$$\begin{aligned} \Sigma_{XX} = \text{E} [(X - \mu)(X - \mu)^T] &= \begin{bmatrix} \text{E} [(X_1 - \text{E}[X_1])(X_1 - \text{E}[X_1])] & \dots & \text{E} [(X_1 - \text{E}[X_1])(X_n - \text{E}[X_n])] \\ \vdots & \ddots & \vdots \\ \text{E} [(X_n - \text{E}[X_n])(X_1 - \text{E}[X_1])] & \dots & \text{E} [(X_n - \text{E}[X_n])(X_n - \text{E}[X_n])] \end{bmatrix} \\ &= \begin{bmatrix} \text{Cov}(X_1, X_1) & \dots & \text{Cov}(X_1, X_n) \\ \vdots & \ddots & \vdots \\ \text{Cov}(X_n, X_1) & \dots & \text{Cov}(X_n, X_n) \end{bmatrix} \end{aligned}$$

- Above taken from [link](#) and [link](#). See the sample covariance [here](#). **Note:** Cov operation is symmetric.

# Discrete Univariate Distribution – Uniform

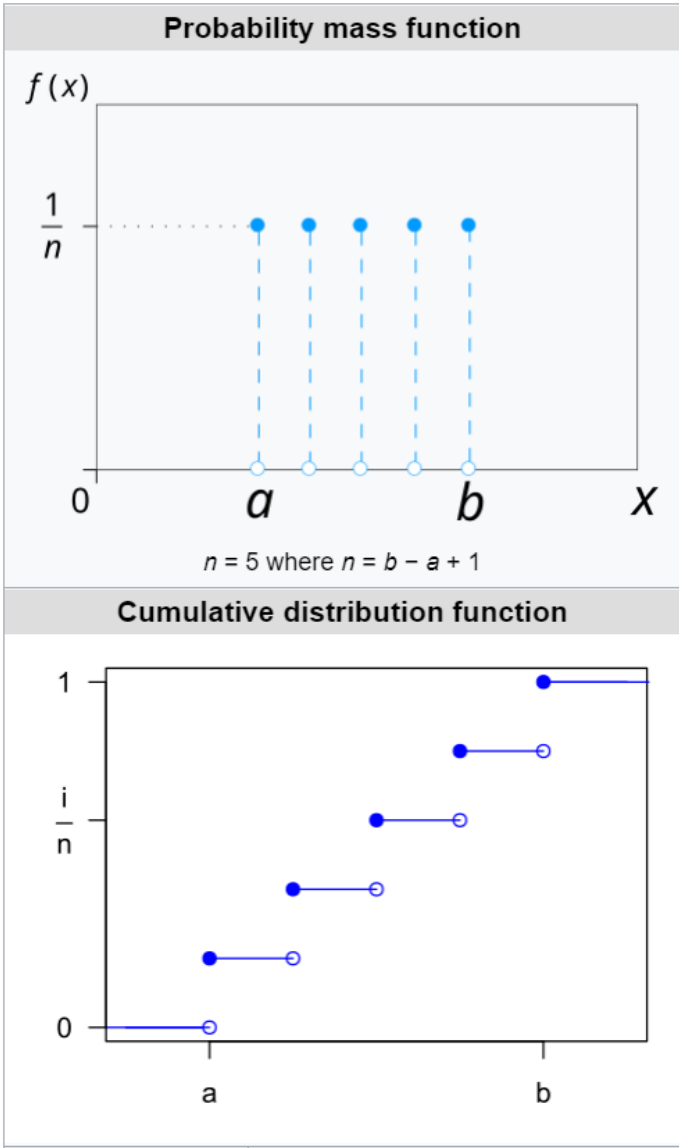
$$E[X] = \sum_x xp(x)$$

$$Var[X] = \sum_x (x - \mu)^2 p(x)$$

- $X \sim U(a, b)$  says that  $X$  is uniformly distributed in range  $[a, b]$ , where it takes on *integer* values with *equal* probability.

|            |                                                    |
|------------|----------------------------------------------------|
| Parameters | $a, b$ integers with $b \geq a$<br>$n = b - a + 1$ |
| Support    | $k \in \{a, a + 1, \dots, b - 1, b\}$              |
| PMF        | $\frac{1}{n}$                                      |
| CDF        | $\frac{\lfloor k \rfloor - a + 1}{n}$              |
| Mean       | $\frac{a + b}{2}$                                  |
| Median     | $\frac{a + b}{2}$                                  |
| Mode       | N/A                                                |
| Variance   | $\frac{(b - a + 1)^2 - 1}{12}$                     |

From [link](#)



# Discrete Univariate Distribution – Bernoulli

$$E[X] = \sum_x xp(x) \qquad Var[X] = \sum_x (x - \mu)^2 p(x)$$

- $X \sim \text{Bernoulli}(p)$  says that  $X$  is a two-outcome process where  $\Pr(X = 1) = p$
- Useful for modelling many real-world two-outcome processes (also called trials).
- Flipping coin, outcome head/tail.
- Outcome of Pass/Fail exam.
- Machine learning task where output is binary label.

|            |                                                                                                                |
|------------|----------------------------------------------------------------------------------------------------------------|
| Parameters | $0 \leq p \leq 1$<br>$q = 1 - p$                                                                               |
| Support    | $k \in \{0, 1\}$                                                                                               |
| PMF        | $\begin{cases} q = 1 - p & \text{if } k = 0 \\ p & \text{if } k = 1 \end{cases}$                               |
| CDF        | $\begin{cases} 0 & \text{if } k < 0 \\ 1 - p & \text{if } 0 \leq k < 1 \\ 1 & \text{if } k \geq 1 \end{cases}$ |
| Mean       | $p$                                                                                                            |
| Median     | $\begin{cases} 0 & \text{if } p < 1/2 \\ [0, 1] & \text{if } p = 1/2 \\ 1 & \text{if } p > 1/2 \end{cases}$    |
| Mode       | $\begin{cases} 0 & \text{if } p < 1/2 \\ 0, 1 & \text{if } p = 1/2 \\ 1 & \text{if } p > 1/2 \end{cases}$      |
| Variance   | $p(1 - p) = pq$                                                                                                |

From [link](#)

# Discrete Univariate Distribution – Binomial

$$E[X] = \sum_x xp(x)$$

$$Var[X] = \sum_x (x - \mu)^2 p(x)$$

- $X \sim Bin(n, p)$  says that  $X$  is number of successes in  $n$  independent trials, where each trial is Bernoulli process.
- A single trial has two outcomes where probability of success is  $p$
- Probability that we get 4 tails in 10 coin flips.
- Probability that 6 students pass out of 10 examinees.
- We may use this distribution later on when talking about model error.

From [link](#)

|            |                                                                                                                     |
|------------|---------------------------------------------------------------------------------------------------------------------|
| Parameters | $n \in \{0, 1, 2, \dots\}$ – number of trials<br>$p \in [0, 1]$ – success probability for each trial<br>$q = 1 - p$ |
| Support    | $k \in \{0, 1, \dots, n\}$ – number of successes                                                                    |
| PMF        | $\binom{n}{k} p^k q^{n-k}$                                                                                          |
| CDF        | $I_q(n - \lfloor k \rfloor, 1 + \lfloor k \rfloor)$ (the <a href="#">regularized incomplete beta function</a> )     |
| Mean       | $np$                                                                                                                |
| Median     | $\lfloor np \rfloor$ or $\lceil np \rceil$                                                                          |
| Mode       | $\lfloor (n + 1)p \rfloor$ or $\lceil (n + 1)p \rceil - 1$                                                          |
| Variance   | $npq = np(1 - p)$                                                                                                   |

# Discrete Multivariate Distribution – Categorical

$$E[X] = \sum_x xp(x) \qquad Var[X] = \sum_x (x - \mu)^2 p(x)$$

From [link](#)

- $X \sim Cat([p_1, \dots, p_k])$  says that  $X$  is a  $k$  outcome process.
- Technically it is **multivariate** since  $X \in \{e_1, \dots, e_k\}$  where:  
 $e_1 = [1, 0, \dots, 0], e_2 = [0, 1, \dots, 0], \dots, e_k = [0, 0, \dots, 1]$   
 $e_i \in \{0, 1\}^k$  are called the elementary or unit vectors.
- Useful for modelling many real-world multiple outcome processes.
- Machine learning task where output is  $k$ -class label.

|                   |                                                                                                                                                                                            |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Parameters</b> | $k > 0$ number of categories ( <a href="#">integer</a> )<br>$p_1, \dots, p_k$ event probabilities<br>( $p_i \geq 0, \Sigma p_i = 1$ )                                                      |
| <b>Support</b>    | $x \in \{1, \dots, k\}$                                                                                                                                                                    |
| <b>PMF</b>        | (1) $p(x = i) = p_i$<br>(2) $p(x) = p_1^{[x=1]} \dots p_k^{[x=k]}$<br>(3) $p(x) = [x = 1] \cdot p_1 + \dots + [x = k] \cdot p_k$<br>where $[x = i]$ is the <a href="#">Iverson bracket</a> |
| <b>Mode</b>       | $i$ such that $p_i = \max(p_1, \dots, p_k)$                                                                                                                                                |

$E[X] = [p_1, p_2, \dots, p_k]$  see derivation [here](#).

$Cov[X] = dig(p) - pp^T$  see derivation [here](#).

# Discrete Multivariate Distribution – Multinomial

$$E[X] = \sum_x xp(x) \qquad Var[X] = \sum_x (x - \mu)^2 p(x)$$

- $X \sim Multinomial(n, [p_1, \dots, p_k])$  says that  $X$  is vector of number of times each outcome occurred in  $n$  independent trials, where each trial is independent.

|            |                                                                                                                                                                     |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Parameters | $n > 0$ number of trials (integer)<br>$k > 0$ number of mutually exclusive events (integer)<br>$p_1, \dots, p_k$ event probabilities, where $p_1 + \dots + p_k = 1$ |
| Support    | $\left\{ (x_1, \dots, x_k) \mid \sum_{i=1}^k x_i = n, x_i \geq 0 \ (i = 1, \dots, k) \right\}$                                                                      |
| PMF        | $\frac{n!}{x_1! \dots x_k!} p_1^{x_1} \dots p_k^{x_k}$                                                                                                              |

From [link](#)

$E[X] = [np_1, np_2, \dots, np_k]$  see derivation [here](#).

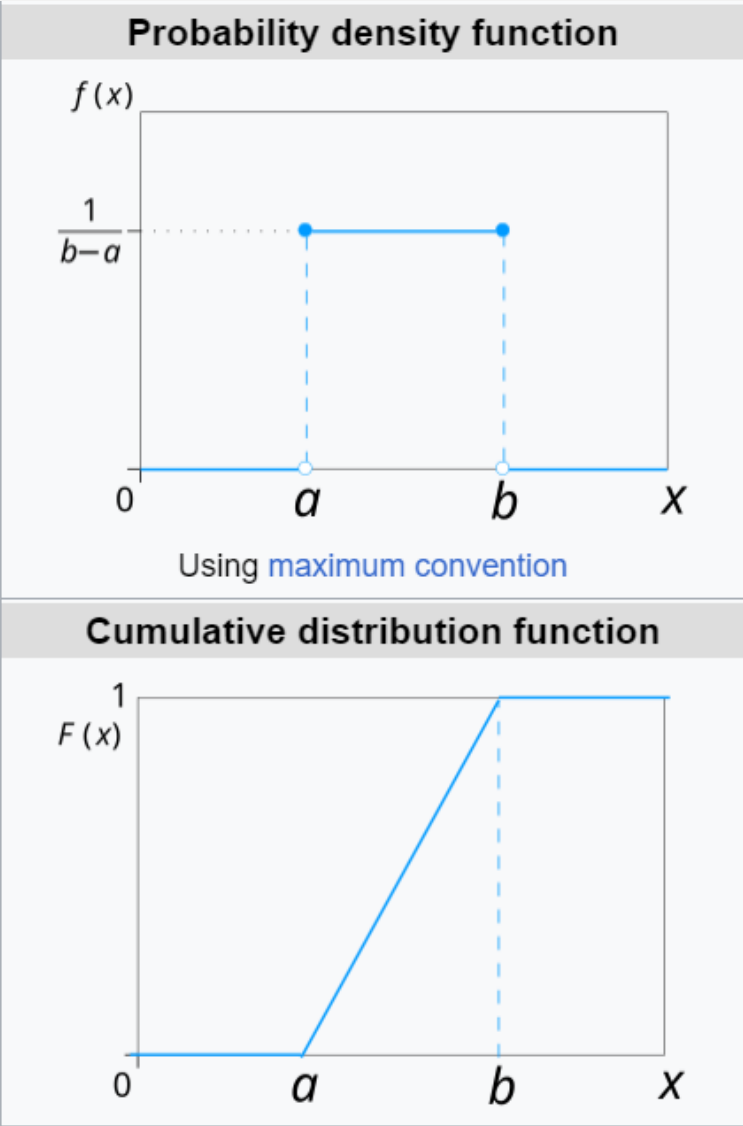
$Cov[X] = n(dig(p) - pp^T)$  see derivation [here](#).

# Continuous Univariate Distribution – Uniform

- $X \sim U(a, b)$  says that  $X$  is uniformly distributed in range  $[a, b]$  of real values.

|            |                                                                                                                          |
|------------|--------------------------------------------------------------------------------------------------------------------------|
| Parameters | $-\infty < a < b < \infty$                                                                                               |
| Support    | $[a, b]$                                                                                                                 |
| PDF        | $\begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$                             |
| CDF        | $\begin{cases} 0 & \text{for } x < a \\ \frac{x-a}{b-a} & \text{for } x \in [a, b] \\ 1 & \text{for } x > b \end{cases}$ |
| Mean       | $\frac{1}{2}(a + b)$                                                                                                     |
| Median     | $\frac{1}{2}(a + b)$                                                                                                     |
| Mode       | any value in $(a, b)$                                                                                                    |
| Variance   | $\frac{1}{12}(b - a)^2$                                                                                                  |

From [link](#)



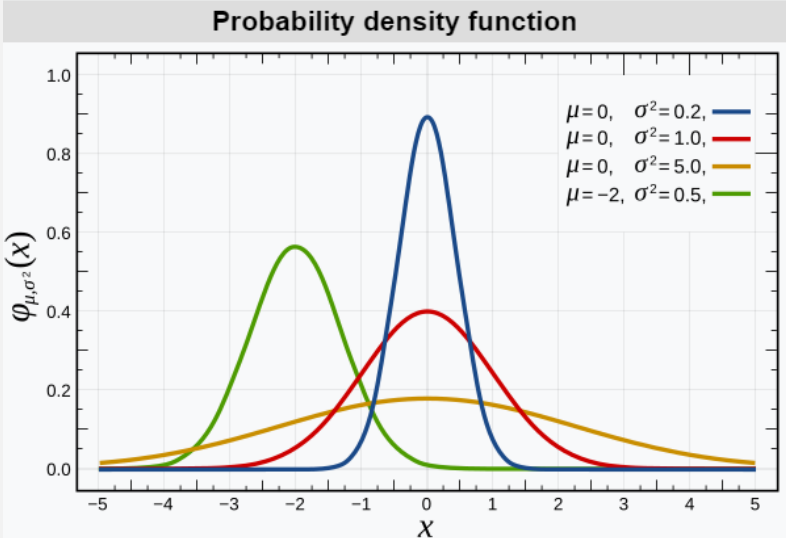
# Continuous Univariate Distribution – Normal (Gaussian)

- $X \sim \mathcal{N}(\mu, \sigma^2)$  says that  $X$  is Normal or Gaussian distributed.
- Used to model many real-valued processes: exam scores, salaries, etc.
- In machine learning, used to model continuous output label.

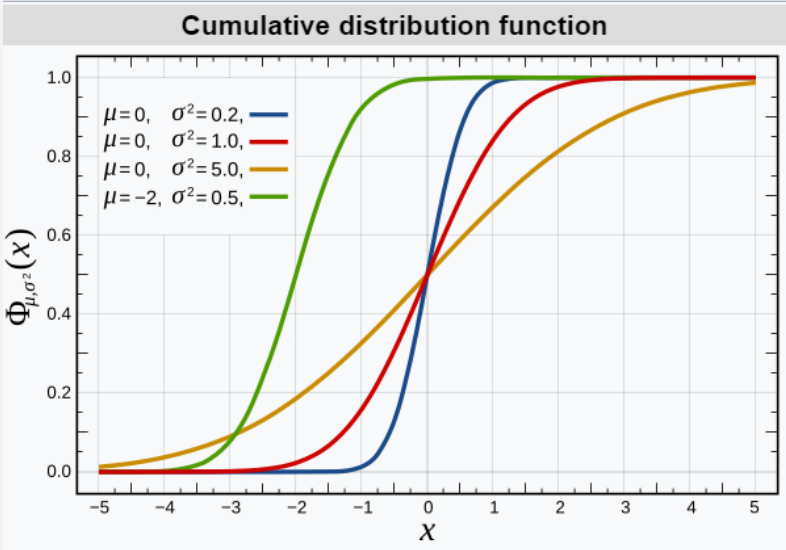
Standard normal is special case  $\mathcal{N}(0,1)$

|            |                                                                                                                                      |
|------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Notation   | $\mathcal{N}(\mu, \sigma^2)$                                                                                                         |
| Parameters | $\mu \in \mathbb{R}$ = mean (location)<br>$\sigma^2 \in \mathbb{R}_{>0}$ = variance (squared scale)                                  |
| Support    | $x \in \mathbb{R}$                                                                                                                   |
| PDF        | $\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$                                                                     |
| CDF        | $\Phi\left(\frac{x-\mu}{\sigma}\right) = \frac{1}{2} \left[ 1 + \operatorname{erf}\left(\frac{x-\mu}{\sigma\sqrt{2}}\right) \right]$ |
| Quantile   | $\mu + \sigma\sqrt{2} \operatorname{erf}^{-1}(2p - 1)$                                                                               |
| Mean       | $\mu$                                                                                                                                |
| Median     | $\mu$                                                                                                                                |
| Mode       | $\mu$                                                                                                                                |
| Variance   | $\sigma^2$                                                                                                                           |

From [link](#)



The red curve is the *standard normal distribution*.



# Continuous Multivariate Distribution – Multivariate Normal (Gaussian)

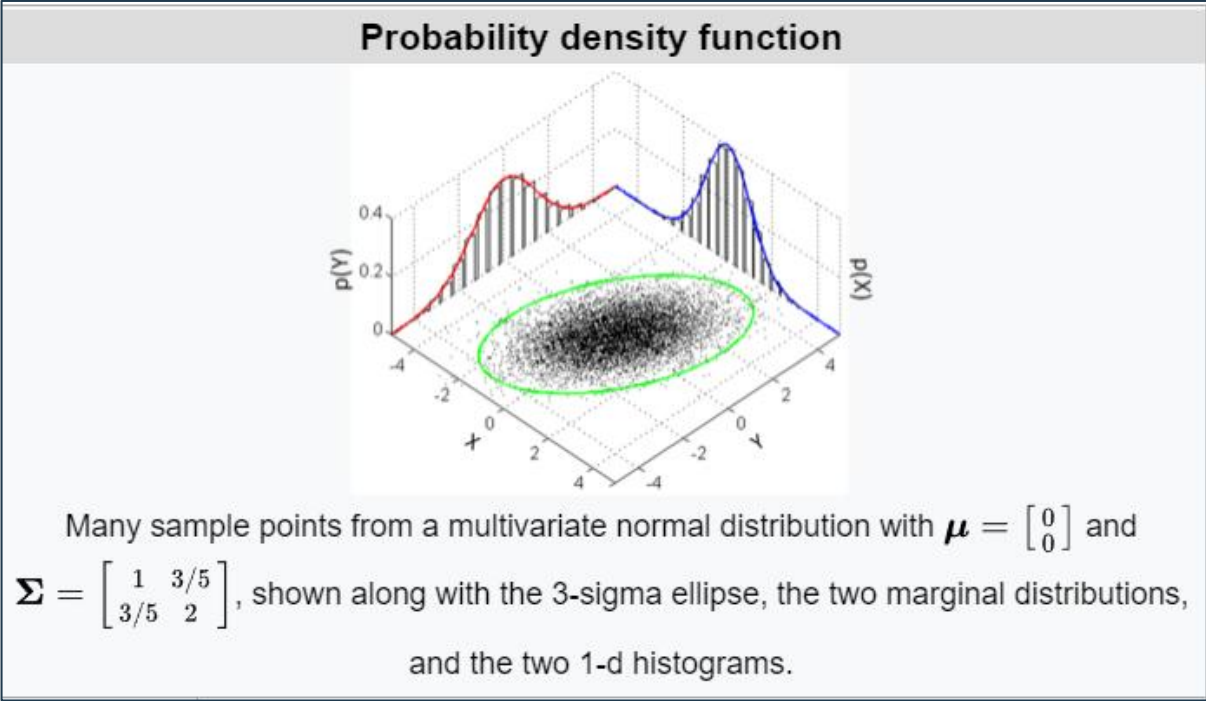
Let  $X$  be an  $n \times 1$  random vector. Then,  $X$  is said to be multivariate normally distributed with mean  $\mu$  and covariance  $\Sigma$ . Denoted as  $X \sim \mathcal{N}(\mu, \Sigma)$ , if and only if its probability density function is given by

$$\mathcal{N}(x; \mu, \Sigma) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} \cdot \exp \left[ -\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right]$$

where  $\mu$  is an  $n \times 1$  real vector and  $\Sigma$  is an  $n \times n$  positive definite matrix.

From [link](#)

|            |                                                                                                                                                            |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Notation   | $\mathcal{N}(\mu, \Sigma)$                                                                                                                                 |
| Parameters | $\mu \in \mathbb{R}^k$ — location<br>$\Sigma \in \mathbb{R}^{k \times k}$ — covariance (positive semi-definite matrix)                                     |
| Support    | $x \in \mu + \text{span}(\Sigma) \subseteq \mathbb{R}^k$                                                                                                   |
| PDF        | $(2\pi)^{-k/2} \det(\Sigma)^{-1/2} \exp \left( -\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right)$ ,<br>exists only when $\Sigma$ is positive-definite |
| Mean       | $\mu$                                                                                                                                                      |
| Mode       | $\mu$                                                                                                                                                      |
| Variance   | $\Sigma$                                                                                                                                                   |



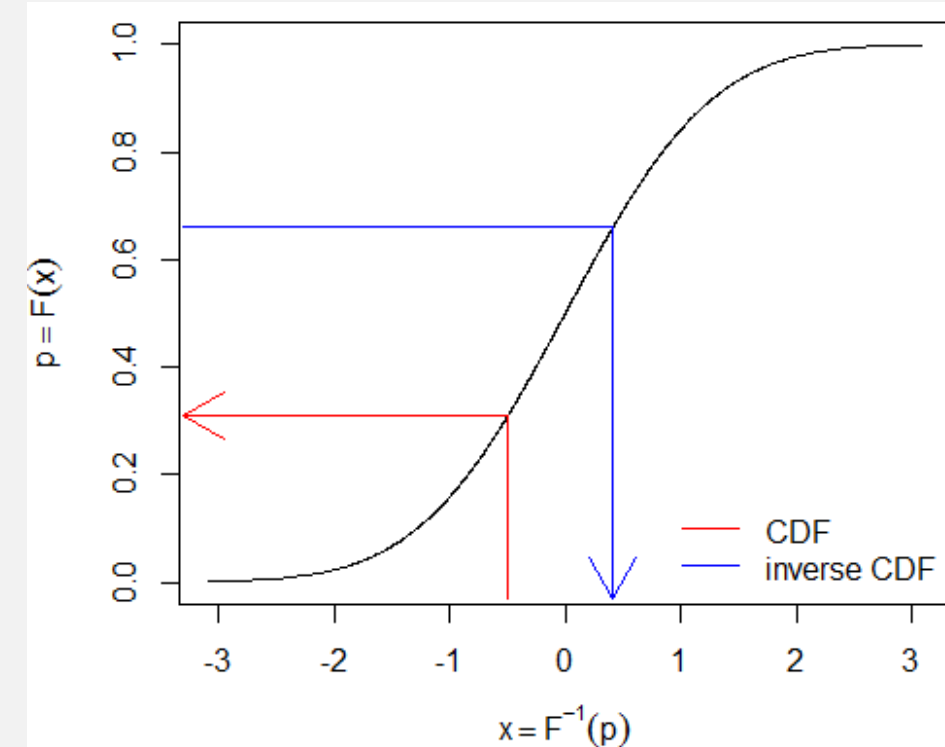
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# Sampling

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# How to sample from a distribution?

- So how is sampling done? Common technique is the inverse transform (also called Smirnov transform).
- Given a random variable that follows a distribution with known CDF  $F(x) = \Pr(X \leq x)$ , we can sample values for that variable.
- The **inverse** of the CDF called the quantile function is given by  $F^{-1}(p) = \inf\{x \mid F(x) \geq p\}$  where  $p \in [0,1]$ . Meaning, it is the smallest value among the set of values of  $x$  for which  $F(x) \geq p$ .
- Consider the standard normal distribution  $Normal(0, 1)$ . The figure shows its CDF curve.
- $F^{-1}(0.67) = \inf\{x \mid F(x) \geq 0.67\}$  get the smallest  $x$  in this set.
- This is approximately  $x \approx 0.4399$



# How to sample from a distribution?

- For many distributions,  $F^{-1}$  cannot be computed *exactly*, but can be approximated well.
- To sample  $X \sim F(x)$  using the inverse transform technique, do:
  1. Sample a value  $u \sim \text{Unif}(0, 1)$ . This  $u$  corresponds to the probability  $p$  above.
  2. See where  $u$  falls in the range  $[0, 1]$ . Use  $x = F^{-1}(u)$ ; alternatively, get the smallest  $x$  for which  $F(x) \geq p$
- $F^{-1}(u)$  has same distribution as  $X \sim F(x)$ . For those interested see intuition and proofs [here](#).
- Discussion: how is  $u \sim \text{Unif}(0,1)$  sampled?
- Discussion: how to sample  $n$  samples of  $X \sim D$ .
- For a discrete distribution, this can be done easily using bins (example on next slides).

# How to sample from a discrete distribution – Categorical Example

- To sample a discrete random variable C:
  - Step 1: Take the outcomes of C and their probabilities, and associate each outcome with a bin in the range [0, 1]. See how this is done for the example given below.
  - Step 2: Sample  $u \sim Unif(0,1)$ . You can use `np.random.uniform()`.
  - Step 3: See where the value of  $u$  falls into the intervals defined in step 1. Grab the outcome that  $u$  falls into

Example:

| C     | P(C) |
|-------|------|
| red   | 0.6  |
| green | 0.1  |
| blue  | 0.3  |

$0.0 \leq u < 0.6, \rightarrow C = \text{red}$   
 $0.6 \leq u < 0.7, \rightarrow C = \text{green}$   
 $0.7 \leq u < 1.0, \rightarrow C = \text{blue}$



Examples of sampling outcomes for C:

- If `np.random.uniform()` returns  $u = 0.83$ , then what is sample for C?  $C = \text{blue}$
- If `np.random.uniform()` returns  $u = 0.65$ , then what is sample for C?  $C = \text{green}$
- If `np.random.uniform()` returns  $u = 0.1$ , then what is sample for C?  $C = \text{red}$

# How to sample from continuous – Normal Example

$$Z = \frac{X - \mu}{\sigma}$$

- To sample a continuous random variable  $C$ :
  - Step 1: Sample  $u \sim \text{Unif}(0,1)$ . You can use `np.random.uniform()`.
  - Step 2: Use  $x = F^{-1}(u)$ ; alternatively, get the smallest  $x$  for which  $F(x) \geq p$
- For normal distribution, no closed form. But in practice approximations are used.
- Straightforward way: after sampling  $u$ , we can use [z-table](#) of standard normal distribution to get z-score. With z-score, we convert it to  $\mathcal{N}(\mu, \sigma^2)$  as desired.
- Example: sample from  $X \sim \mathcal{N}(100, 25)$ :
  - Step 1: Sample  $u \sim \text{Unif}(0,1)$ . Let's say we got  $u = 0.95$
  - Step 2: consult z-table to find smallest  $z$  for which  $F(z) \geq 0.95$ ; this would  $z \approx 1.65$
  - Step 3: convert to desired normal distribution  $x = z\sigma + \mu = 1.65 * 5 + 100 = 108.25$
- In practice, other fast approximations are used.
- You would not need to implement this process yourself. In python, you can use numpy's [np.random.normal](#). Numpy also supports sampling from other distributions.

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# Parameter Estimation

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# Parameter Estimation Overview

- Suppose that you are given the following information:
  - $X \sim D$  saying that  $X$  is a scalar random variable following some distribution  $D$ .
  - $n$  samples of  $X$ . Typically, we denote these samples as  $x_1, x_2, \dots, x_n$
  - We typically assume that these samples are independent and identically distributed (iid).
- **Parametric Estimation**: you *assume* the form of the distribution  $D$  (pmf/cdf), but do not know the exact parameter values of  $D$ . We'll denote the parameters as  $\theta$ .
  - Typically, in practice you are not given what distribution  $D$  is. You can sometimes visualize the data to get some ideas.
  - Once you assume a distribution for  $D$ , the parameters become clear, but their *values* are unknown.
  - You can estimate the values of  $\theta$  using the  $n$  samples. Typically, **estimators** are denoted with a hat, i.e.  $\hat{\theta}$
  - The estimation method you use can be good or bad. Typically, we judge an estimator by the **bias** and **variance**.
  - With a good estimator, it converges to the true  $\theta$  as  $n \rightarrow \infty$ .
  - Some ML algorithms use parametric estimation (i.e. assumes the form of the distribution).
- **Non-parametric Estimation**: we do not assume any form on  $D$ .
  - You use the  $n$  samples to construct a *free-form* distribution based directly on the data.
  - Non-parametric does not mean no parameters are used, but that the number and nature of the parameters are flexible and not fixed in advance.
  - Examples: histogram, kernel-density, k-nearest neighbor ML algorithm.

# Maximum Likelihood Estimation (MLE)

- Let  $D_\theta$  be some assumed distribution with parameter  $\theta$  (value is unknown).
- Let  $S = (x_1, x_2, \dots, x_n)$  be  $n$  samples that are iid from  $D_\theta$
- MLE is a common method for estimating parameters of an assumed distribution from samples. Commonly called the MLE estimator.
- Briefly, the MLE estimator of the unknown parameter is the maximizer of the **likelihood** of the samples.
- What is the likelihood? We can use the distribution's probability function (pmf/pdf).

- Here is the formulation:

$$\begin{aligned}\hat{\theta} &= \operatorname{argmax}_{\theta} \mathcal{L}(S; D_\theta) \\ &= \operatorname{argmax}_{\theta} \operatorname{Prob}(S) \\ &= \operatorname{argmax}_{\theta} \operatorname{Prob}(x_1, x_2, \dots, x_n)\end{aligned}$$

- How to solve this? Depends on the form of the likelihood function we use. We borrow techniques from optimization field.

## Some useful rules [1]

- Finding the maximum of a function  $g(w)$  is the same as finding the maximum of  $\log g(w)$ .

$$\begin{aligned}w^* &= \arg \max_{w \in \mathbb{R}^{n+1}} g(w) \\&= \arg \max_{w \in \mathbb{R}^{n+1}} \log(g(w))\end{aligned}$$

- Finding the maximum of a function  $g(w)$  is the same as finding the minimum of  $-g(w)$ .

$$\begin{aligned}w^* &= \arg \max_{w \in \mathbb{R}^{n+1}} g(w) \\&= \arg \min_{w \in \mathbb{R}^{n+1}} -g(w)\end{aligned}$$

## Some useful rules [2]

- The log of products can be changed to sum of logs. If  $a$  and  $b$  are scalars, then:

$$\log(ab) = \log(a) + \log(b)$$

- The log of quotients can be changed to difference of logs. If  $a$  and  $b$  are scalars, then:

$$\log\left(\frac{a}{b}\right) = \log(a) - \log(b)$$

- Power rule of log. If  $a$  and  $b$  are scalars, then:

$$\log(a^b) = b \log(a)$$

# Parametric Estimation Example – Bernoulli [1]

- Suppose we have  $n$  samples  $x_1, x_2, \dots, x_n$  where  $x_i \in \{0, 1\}$ . A reasonable assumption is that these samples are independent and identically distributed following *Bernoulli*( $p$ )
- We do not know  $p$ , how can we estimate it from the samples?
- A reasonable estimator is sample mean:  $\hat{p} = \frac{1}{n} \sum_i x_i$
- This is the MLE estimator.
- This estimator is **unbiased**.
- See remaining derivation next slide.

$$\begin{aligned}\hat{p} &= \operatorname{argmax}_p \operatorname{Prob}(x_1, x_2, \dots, x_n) \\ &= \operatorname{argmax}_p \prod_i \operatorname{Prob}(x_i) \\ &= \operatorname{argmax}_p \prod_i p^{x_i} (1 - p)^{1 - x_i} \\ &= \operatorname{argmax}_p \log \left[ \prod_i p^{x_i} (1 - p)^{1 - x_i} \right] \\ &= \operatorname{argmax}_p \sum_i \log \left[ p^{x_i} (1 - p)^{1 - x_i} \right] \\ &= \operatorname{argmax}_p \sum_i x_i \log(p) + \sum_i (1 - x_i) \log(1 - p)\end{aligned}$$

## Parametric Estimation Example – Bernoulli [2]

$$\begin{aligned}\hat{p} &= \operatorname{argmax}_p \operatorname{Prob}(x_1, x_2, \dots, x_n) \\ &= \operatorname{argmax}_p \sum_i x_i \log(p) + \sum_i (1 - x_i) \log(1 - p) \\ &= \operatorname{argmin}_p - \sum_i x_i \log(p) - \sum_i (1 - x_i) \log(1 - p)\end{aligned}$$

- One way to find min of a function is to set its derivative to zero, and solve for the parameter.

$$\frac{\partial}{\partial p} \left[ - \sum_i x_i \log(p) - \sum_i (1 - x_i) \log(1 - p) \right] = - \frac{\sum_i x_i}{p} + \frac{\sum_i (1 - x_i)}{1 - p}$$

- Now set to zero, and solve for the parameters (i.e p in this case).

$$\begin{aligned}- \frac{\sum_i x_i}{p} + \frac{\sum_i (1 - x_i)}{1 - p} = 0 &\implies \frac{\sum_i (1 - x_i)}{1 - p} = \frac{\sum_i x_i}{p} \implies p \sum_i 1 - p \sum_i x_i = \sum_i x_i - p \sum_i x_i \\ &\implies pn = \sum_i x_i \implies p = \frac{\sum_i x_i}{n}\end{aligned}$$

# Estimator Bias and Variance

- Let  $X \sim D_\theta$ . Let  $\theta$  denote true parameter (unknown value) and let  $\hat{\theta}$  denote the estimator.
- Note that  $\hat{\theta}$  is typically a *method* that computes a value from sample data.
- The estimator bias is given by:

$$\text{Bias}(\hat{\theta}, \theta) = \text{E}[\hat{\theta}] - \theta$$

- The estimator variance is given by:

$$\text{Var}(\hat{\theta}) = \text{E} \left[ \left( \hat{\theta} - \text{E}[\hat{\theta}] \right)^2 \right]$$

- The estimator mean-squared-error (MSE) is given by bias and variance:

$$\text{MSE}(\hat{\theta}) = \text{E} \left[ \left( \hat{\theta} - \theta \right)^2 \right] = \text{Bias}(\hat{\theta}, \theta)^2 + \text{Var}(\hat{\theta})$$

- An estimator with zero bias is said to be *unbiased* estimator.
- What we'd like is an estimator with low bias and low variance. However, as we'll talk about later in the course, *typically* reducing bias leads to increased variance, and reducing variance leads to increasing bias.

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# Statistical Modelling and Probabilistic Inference

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# Statistical Modelling of Environments

- Now that we covered many tools from probability, we can look at examples of developing probability models.
- In general, to deal with complex environments with uncertainty, modelling the environment as a probabilistic process helps in decision making and managing risk.
- This involves defining the properties of the environment as random variables following some distribution.
- For example, consider you want to model relationship between weather and temperature as a probabilistic process.
- We have seen a simple model of this earlier in these slides. This model has two variables T and W.
- Discussion: which common distribution is this?
- Let's say we do not make assumptions on the relationship between these two variables, so, we need to define the full-joint probabilities.
- The full-joint table provides complete information with no assumptions. However, there are time complexity issues with this.

$P(T,W)$

|         |        | Temperature |      |
|---------|--------|-------------|------|
|         |        | hot         | cold |
| Weather | sun    | 0.45        | 0.15 |
|         | rain   | 0.02        | 0.08 |
|         | fog    | 0.03        | 0.27 |
|         | meteor | 0.00        | 0.00 |

# Model with no assumptions among variables

- Consider an environment with  $n$  random variables  $x_1, x_2, \dots, x_n$
- No assumptions are made on the relationship among the variables.
- This means when modelling the probability model, it is given by the full-joint:

$$Prob(x_1, x_2, \dots, x_n)$$

- Let's say that each variable is discrete and can take on  $d$  values, how many probability entries to store?  $O(d^n)$
- So, this is space complexity to store the entries. What about time complexity of answering a query?
- **Probabilistic inference**: compute the probability of a desired event (**query**) using a **probability model**
  - Typically for **query variables given evidence**;  $P(Q1, Q2, \dots \mid \text{evidence1, evidence2, } \dots)$
  - E.g.,  $\text{Prob}(\text{airport on time} \mid \text{no accidents}) = 0.90$
  - Represents agent's beliefs given the evidence.
  - Note we can also do inference for query variables without evidence  $P(Q1, Q2, \dots)$

# Inference by **enumeration** using Full Joint Distribution Table

- Given the full joint distribution table  $P(X_1, \dots, X_n)$  we can perform inference by enumeration.
- We want to answer a query:  $P(q \mid e)$
- Partition the variables  $X_1, \dots, X_n$  into three sets as follows:
  - Query** variables:  $q$
  - Evidence** variables:  $e$
  - Hidden** variables (non-query variables):  $H$

| $X_1$    | $X_2$    | ... | $X_n$    | $P(X_1, \dots, X_n)$ |
|----------|----------|-----|----------|----------------------|
| $X1val1$ | $X2val1$ | ... | $Xnval1$ | $p_1$                |
| $X1val1$ | $X2val1$ | ... | $Xnval2$ | $p_2$                |
| ...      | ...      | ... | ...      | ...                  |
| $X1vald$ | $X2vald$ | ... | $Xnvald$ | $p_n$                |

$$P(q \mid e) = P(q, e) / P(e)$$

- Step 1: Sum out  $H$  from model to get joint of query and evidence. This is summing entries that are consistent with  $q$  and  $e$ .
- Step 2: In the joint-prob table, find entries consistent with the evidence. Then sum their probas.
- Step 3: Compute result

$$P(q \mid e) = P(q, e) / P(e)$$

$$P(q, e) = \sum_h P(q, e, h)$$

$$P(e) = \sum_x P(e, x)$$

# Full-joint inference example

- Use enumeration to compute:  $P(W = sun)$ ?
- This query has no evidence. We proceed by summing out the hidden variables. This is summing entries with (W = sun).

$$P(W = sun) = \sum_{s,t} P(W = sun, S = s, T = t)$$
$$= 0.33 + 0.10 + 0.10 + 0.10 = 0.63$$

| Weather | $P(W)$ |
|---------|--------|
| sun     | 0.63   |
| rain    | 0.29   |
| fog     | 0.08   |
| meteor  | 0.00   |

| Season | Temp | Weather | P    |
|--------|------|---------|------|
| summer | hot  | sun     | 0.33 |
| summer | hot  | rain    | 0.01 |
| summer | hot  | fog     | 0.01 |
| summer | hot  | meteor  | 0.00 |
| summer | cold | sun     | 0.10 |
| summer | cold | rain    | 0.05 |
| summer | cold | fog     | 0.05 |
| summer | cold | meteor  | 0.00 |
| winter | hot  | sun     | 0.10 |
| winter | hot  | rain    | 0.03 |
| winter | hot  | fog     | 0.02 |
| winter | hot  | meteor  | 0.00 |
| winter | cold | sun     | 0.10 |
| winter | cold | rain    | 0.20 |
| winter | cold | fog     | 0.00 |
| winter | cold | meteor  | 0.00 |

# Full-joint inference example

- Use enumeration to compute:  $P(W=\text{rain} \mid \text{Season}=\text{winter})$ ?

$$P(W = \text{rain} \mid S = \text{winter}) = \frac{P(W = \text{rain}, S = \text{winter})}{P(S = \text{winter})}$$
$$= \frac{0.23}{0.45} = 0.51$$

| Weather | $P(W \mid S = \text{winter})$ |
|---------|-------------------------------|
| sun     | 0.45                          |
| rain    | 0.51                          |
| fog     | 0.04                          |
| meteor  | 0.00                          |

| Season | Temp | Weather | P    |
|--------|------|---------|------|
| summer | hot  | sun     | 0.33 |
| summer | hot  | rain    | 0.01 |
| summer | hot  | fog     | 0.01 |
| summer | hot  | meteor  | 0.00 |
| summer | cold | sun     | 0.10 |
| summer | cold | rain    | 0.05 |
| summer | cold | fog     | 0.05 |
| summer | cold | meteor  | 0.00 |
| winter | hot  | sun     | 0.10 |
| winter | hot  | rain    | 0.03 |
| winter | hot  | fog     | 0.02 |
| winter | hot  | meteor  | 0.00 |
| winter | cold | sun     | 0.10 |
| winter | cold | rain    | 0.20 |
| winter | cold | fog     | 0.00 |
| winter | cold | meteor  | 0.00 |

# Full-joint table parameter estimation

- Consider the Season, Temp, Weather model with the given domains.
- How many parameters does this model have?
- What if you don't know the probabilities? How to estimate it?
- In practice, when you use a model, you are assuming that data is generated according to your model. You gather data from the real-world.
- For this particular model, a sample would be a 3-element vector with values for Season, Temp, Weather.
- Say you have collected  $m$  samples for this model.
- How to estimate the parameters?
  - The straightforward method would be  $\hat{p}(s, t, w) = \frac{\# \text{ samples with } (s, t, w)}{m}$
  - Other methods we won't discuss now (maybe later in the course).
- The more samples you have, the better the estimation.

| Season | Temp | Weather | P |
|--------|------|---------|---|
| summer | hot  | sun     | ? |
| summer | hot  | rain    | ? |
| summer | hot  | fog     | ? |
| summer | hot  | meteor  | ? |
| summer | cold | sun     | ? |
| summer | cold | rain    | ? |
| summer | cold | fog     | ? |
| summer | cold | meteor  | ? |
| winter | hot  | sun     | ? |
| winter | hot  | rain    | ? |
| winter | hot  | fog     | ? |
| winter | hot  | meteor  | ? |
| winter | cold | sun     | ? |
| winter | cold | rain    | ? |
| winter | cold | fog     | ? |
| winter | cold | meteor  | ? |

# Full-joint inference complexity

- Suppose we have a probability model with  $n$  variables and  $d$  possible values for each variable.
- With no assumptions, model is the full-joint with all variables.
- Space complexity? How many rows does the joint-probability table have
  - $O(d^n)$  to store the full joint distribution table.
- Worst case time complexity to answer query?
  - In general, for a given query with evidence, we need to do summation over hidden variables.
  - Consider one query variable and no evidence  $P(X_1 = x_1)$
  - Akin to scanning the entire table to find where  $X_1 = x_1$  and sum these entries.
  - In general, can take worst-case  $O(d^n)$  to process and sum over the table entries.
- Need other efficient methods for performing inference.
- To get efficiency, we make assumptions to enforce a structure on the model:
  - Independence assumptions.
  - Conditional Independence assumptions and naïve Bayes assumption.
  - Bayesian networks (depends on specific structure; generally exponential).

# Independence Property

- Two variables X and Y are (absolutely) **independent** if

$$\forall x, y \quad P(X=x, Y=y) = P(X=x) P(Y=y)$$

- I.e., the joint distribution **factors** into a product of two simpler distributions

- Equivalently, via the product rule  $P(x,y) = P(x|y)P(y)$ ,  
 $P(x | y) = P(x)$  or  $P(y | x) = P(y)$

- Note:** this needs to hold for *all* possible values of the variables.

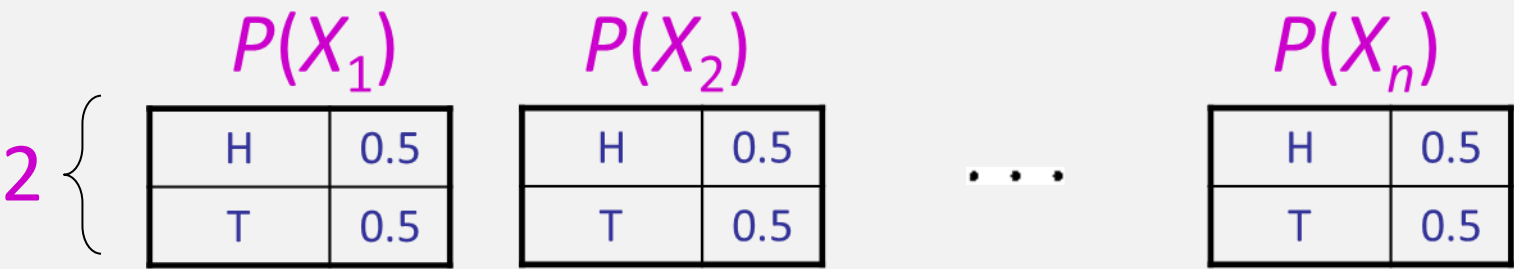
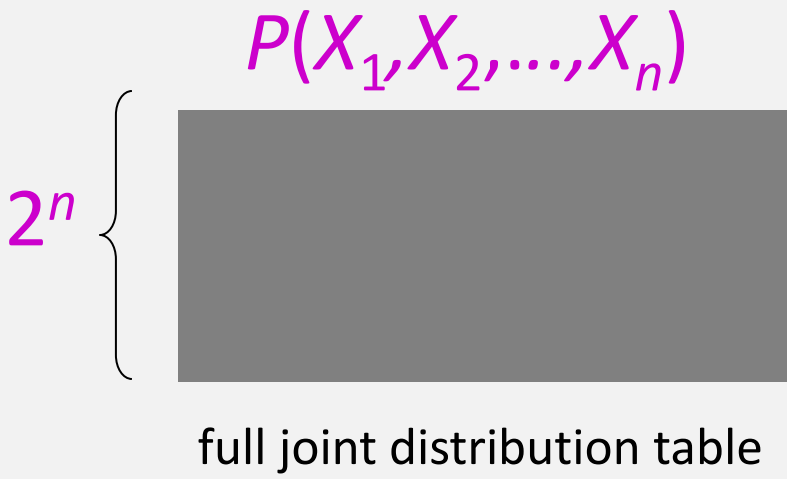
- Example: two dice rolls Roll1 and Roll2 that are independent

- $P(\text{Roll}_1=5, \text{Roll}_2=3) = P(\text{Roll}_1=5) P(\text{Roll}_2=3) = 1/6 \times 1/6 = 1/36$
- $P(\text{Roll}_2=3 | \text{Roll}_1=5) = P(\text{Roll}_2=3)$

# Exploiting Independence

- Suppose we have  $n$  fair **independent** coin flips.
- The full joint distribution table consists of  $n$  variables with 2 values each.
- Using independence, we get:

$$P(X_1, X_2, \dots, X_n) = P(X_1)P(X_2) \dots P(X_n)$$



- For this problem, using independence, our *worst-case* space and time complexity goes from  $2^n$  to  $2n$ .
- Thus, exploiting independence is one way to reduce complexity of inference.

# Conditional Independence and Naïve Bayes

- X is **conditionally independent** of Y given Z if and only if:

$$\forall x, y, z \quad P(x, y \mid z) = P(x \mid z) P(y \mid z)$$

- Also denoted as:  $X \perp Y \mid Z$

- Equivalently, if and only if,

$$\forall x, y, z \quad P(x \mid y, z) = P(x \mid z)$$

$$\forall x, y, z \quad P(y \mid x, z) = P(y \mid z)$$

- Can generalize to more than 2 random variables.
  - E.g., k different symptom variables  $X_1, X_2, \dots, X_K$ , and D = disease, reducing to:
$$P(X_1, X_2, \dots, X_K \mid D) = \prod P(X_i \mid D)$$
  - Also known as **naïve Bayes** assumption: a set of variables are all conditionally independent of each other given D.

## Exploitation comparison: full-joint table vs independence vs conditional independence

- If we had  $N$  Binary Random Variables
  - Full Joint Probability Table (no assumptions)
    - 1 Table with  $2^N$  Rows;  $N = 100$ ,  $2^{100} \approx 10^{30}$
  - Assuming variables are independent of each other:
    - $N$  tables. Total entries  $1 + 1 + \dots + 1 = O(N)$
  - Assuming Naïve Bayes Model (Conditional Independence of  $n - 1$  variables given one variable)
    - $N$  tables. Total entries  $1 + 2 + 2 + \dots + 2 = 1 + 2N = O(N)$
- Thus, exploiting these independence is one way to reduce complexity of inference.
- Another approach is to impose structure on how the variables influence each other using Bayesian networks.

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# Bayesian Networks

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# Probabilistic Graphical Model (PGM)

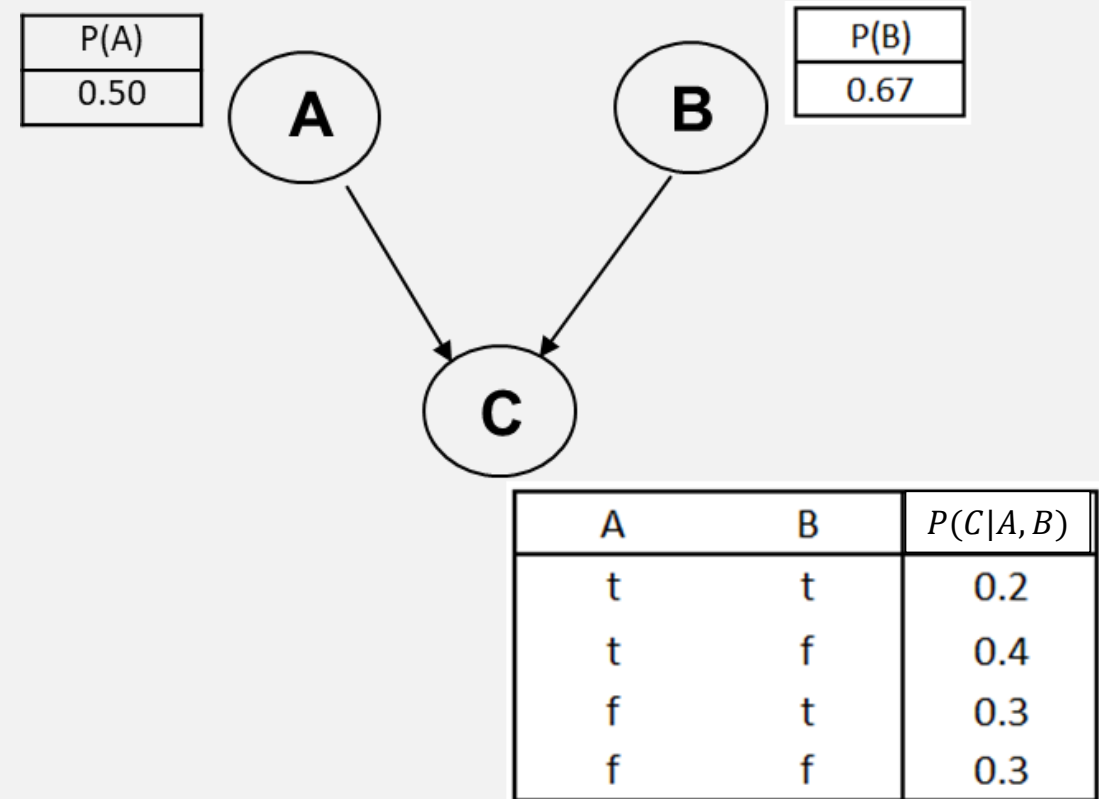
- PGMs: a way to define the structure of variable interaction in a probability model using graphs (nodes and edges)
- Useful for human visualization and understanding of how variables interact, and samples are generated.
- Consider an environment with  $n$  variables  $X_1, X_2, \dots, X_n$ .
  - A PGM  $G_1 = (V_1, E_1)$  for this environment would have the variables as nodes and relationship between them as edges.
  - Another PGM  $G_2 = (V_2, E_2)$  will still have variables as nodes, but edges may be different to represent different interaction.
  - How many possible graphs for  $n$  variables?
- Semantically, the presence of an edge between  $X$  and  $Y$  indicates direct dependence relationship.
- Typically constructed using domain knowledge (handcrafted), but algorithms exist to construct from data.
- Types of PGMs:
  - Undirected PGMs: undirected edges and indicate *correlation* between variables. (for those interested see [link](#))
  - Directed PGMs: directed edges indicate *causality*.
  - Bayesian Networks (Directed Acyclic Graphs): edges are directed, indicate causality, and **no** cycles in graph.
- Our focus will be Bayesian Networks due to its strong ties with ML.

# Bayesian Networks – Overview

- **Bayesian networks**: a technique for describing complex joint distributions (probability models) using simple conditional distributions (causality relationships).
- Idea: Use local causality/conditional independence; the world is composed of many variables, each interacting locally with a few others.
- Provides a way to **compactly** represent the full-joint distribution.
- Bayesian networks are visually represented as a directed graph (nodes and directed edges).

# Bayesian Networks – Syntax

- Bayesian networks are visually represented as a directed graph (nodes and directed edges).
  - Nodes represent random variables.
  - Directed edge represent conditional independence.
- Allows concise view of joint distribution relationships:
  - Nodes and edges show conditional (causal) relationships between variables.
  - Conditional distribution represented as a **conditional probability table** (CPT) giving the distribution over variable  $X$  for each combination of  $Parents(X)$ .

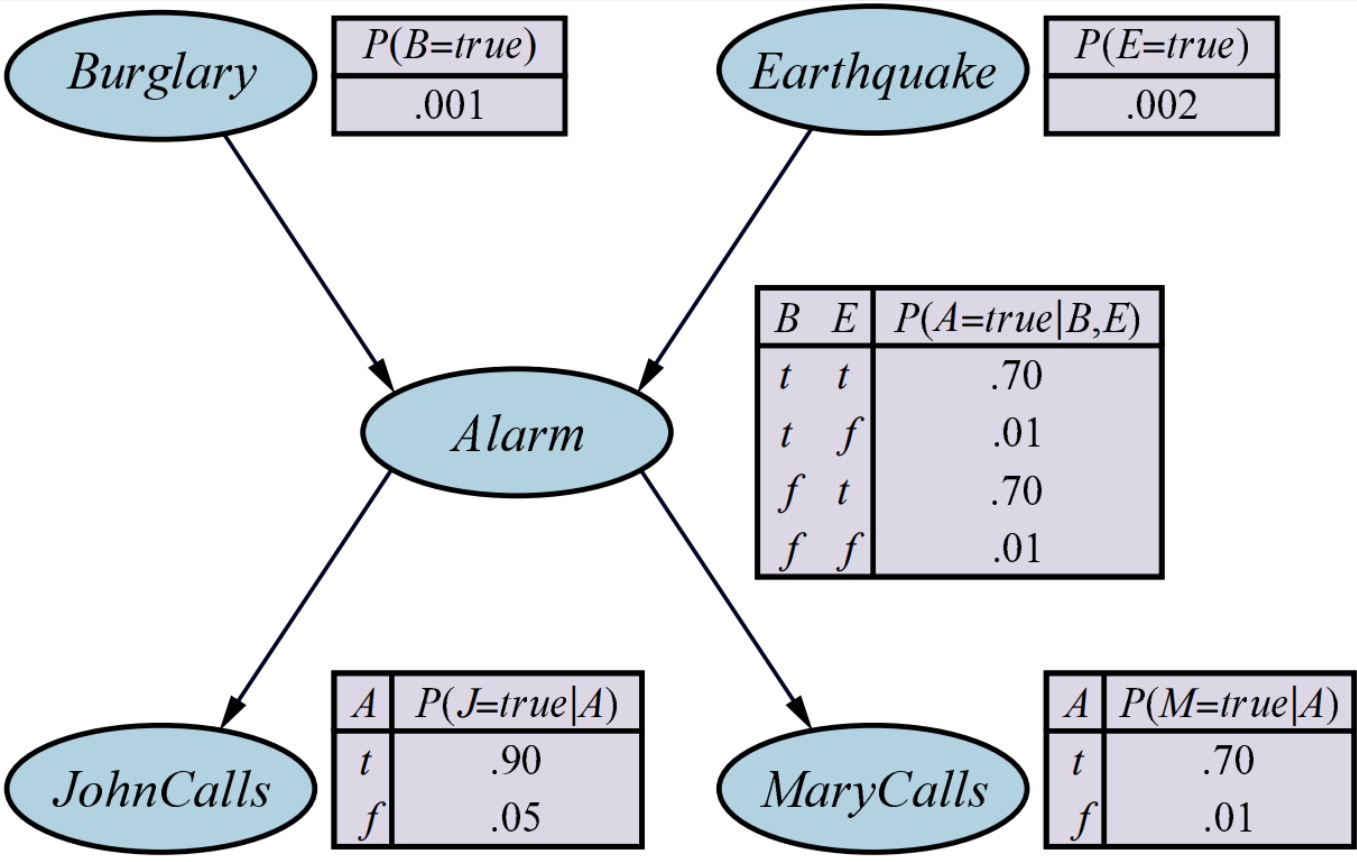


Example of Bayesian network for a model with binary variables A, B, and C. Semantically, the joint probability is given by:

$$P(A, B, C) = P(A)P(B)P(C|A, B)$$

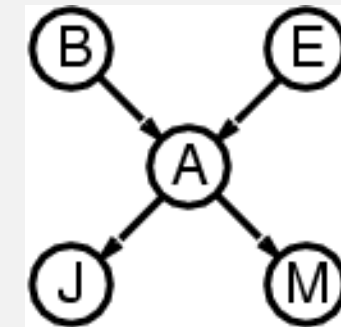
# Example: Burglary Network

- I'm at work, neighbor John calls to say my alarm is ringing, but neighbor Mary doesn't call. Sometimes it's set off by minor earthquakes. Is there a burglar?
- Variables: *Burglary*, *Earthquake*, *Alarm*, *JohnCalls*, *MaryCalls*
- Network topology reflects "causal" knowledge:
  - An alarm caused by burglar and earthquake.
  - MaryCall caused by alarm.
  - JohnCall caused by alarm.
- Number of entries in each CPT? Depends on number of parents
  - Burglar needs 1
  - Earthquake needs 1
  - Alarm needs  $2 \times 2 = 4$
  - JohnCalls needs 2
  - MaryCalls needs 2



# Compactness

- A CPT for binary  $X_i$  with  $k$  binary parents has  $2^k$  rows for the combinations of parent values.
  - Each row requires one entry for  $X_i = \text{true}$
  - If each variable has no more than  $k$  parents, the complete network requires  $O(n2^k)$  entries.
  - I.e., grows linearly with  $n$ , vs.  $O(2^n)$  for the full joint distribution.
- For alarm net,  $1 + 1 + 4 + 2 + 2 = 10$  numbers (vs.  $2^5 - 1 = 31$ )
- For a general BN with
  - $n$  variables
  - max of  $d$  possible values for each variable
  - Max of  $k$  parents for each variable
  - Full joint table has  $O(d^n)$  entries
  - **BN space complexity**  $O(n d^{k+1})$  entries; linear in  $n$



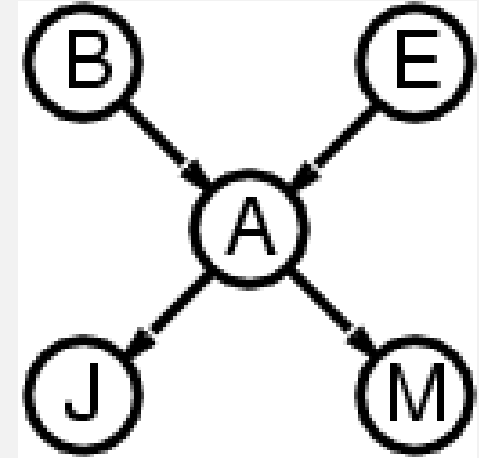
# Bayesian Networks – Semantics

- Bayes nets encode joint distributions as product of conditional distributions on each variable:

$$P(X_1, X_2, \dots, X_n) = \prod_i P(X_i \mid \text{Parents}(X_i))$$

- For alarm net, what is  $P(j \wedge m \wedge a \wedge \neg b \wedge \neg e)$  ?

$$= P(\neg b) P(\neg e) P(a \mid \neg b, \neg e) P(j \mid a) P(m \mid a)$$



Notice that the network gives the order of *sampling*  
First B and E are sampled, then A is sampled given B and E,  
then J is sampled given A, M is sampled given A.

# Pgmpy python package

- [pgmpy](#) python package for Bayesian networks.
- It is a package for Bayesian networks. Many assumptions can be modelled as a Bayesian network.
- You can provide the BN structure yourself or use structure-learning algorithms given a dataset.
- You can provide the probabilities yourself or use estimation algorithms given a dataset.
- Supports Exact inference and Approximate (Sampling Based).

## Navigation

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# Bayesian Networks – Semantics

- Compare BN semantics:

$$P(X_1, X_2, \dots, X_n) = \prod_i P(X_i \mid \text{Parents}(X_i))$$

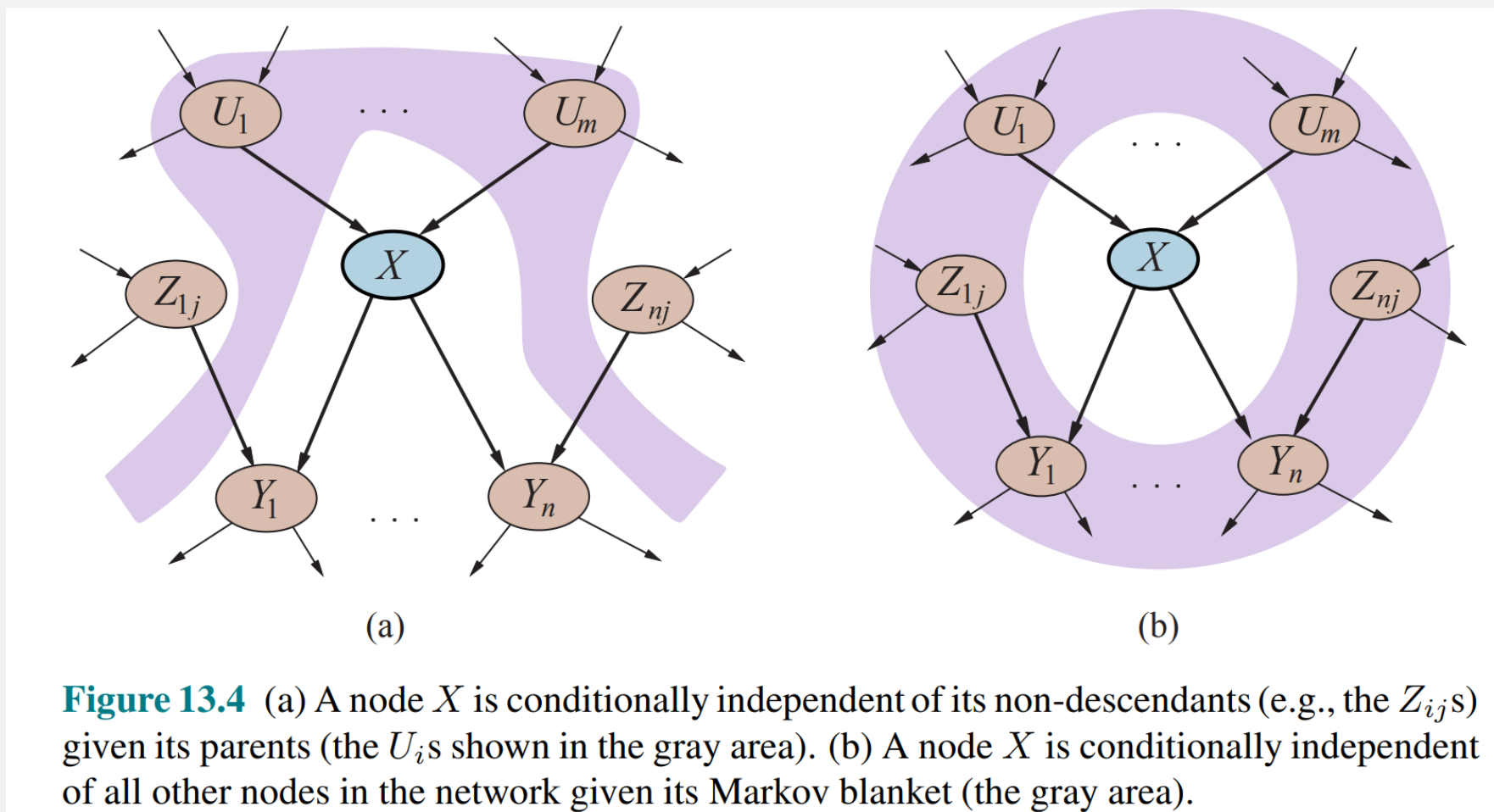
with the chain rule identity:

$$P(X_1, X_2, \dots, X_n) = \prod_i P(X_i \mid X_1, X_2, \dots, X_{i-1})$$

- So, the Bayes net asserts **conditional independences**  $P(X_i \mid X_1, X_2, \dots, X_{i-1}) = P(X_i \mid \text{Parents}(X_i))$

## Conditional independence semantics in BNs

- From the conditional independence in BNs, we get two nice properties.
- “Every variable is conditionally independent of its non-descendants given its parents”
- “a node is conditionally independent of all other nodes in the network, given its parents, children, and children’s parents—that is, given its **Markov blanket**”



# Conditional independence semantics in BNs

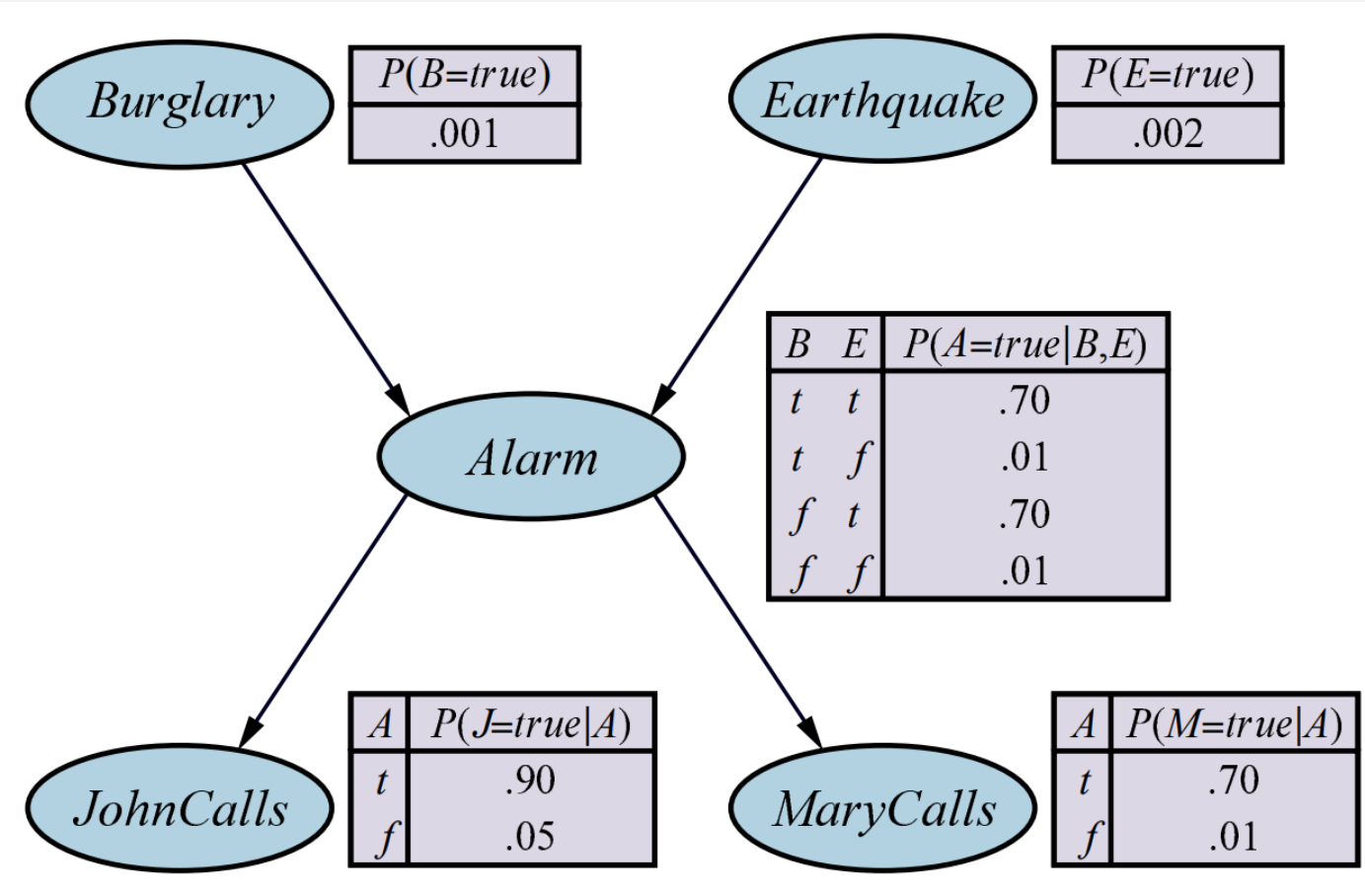
- “Every variable is conditionally independent of its non-descendants given its parents”
- “a node is conditionally independent of all other nodes in the network, given its parents, children, and children’s parents—that is, given its **Markov blanket**”

- JohnCalls* is independent of *Burglary*, *Earthquake*, and *MaryCalls* given *Alarm*.

$$P(J, B, E, M \mid A) = P(J \mid A) P(B, E, M \mid A)$$

- Burglary* is independent of *JohnCalls* and *MaryCalls* given *Alarm* and *Earthquake*.  
(Markov Blanket of Burglary)

$$P(J, B, M \mid A, E) = P(J, M \mid A, E) P(B \mid A, E)$$



“Every variable is conditionally independent of its non-descendants given its parents” Example

What is  $P(x_5, x_6 \mid x_2, x_3)$  ?

$$P(x_5, x_6 \mid x_2, x_3) = \frac{P(x_5, x_6, x_2, x_3)}{P(x_2, x_3)}$$

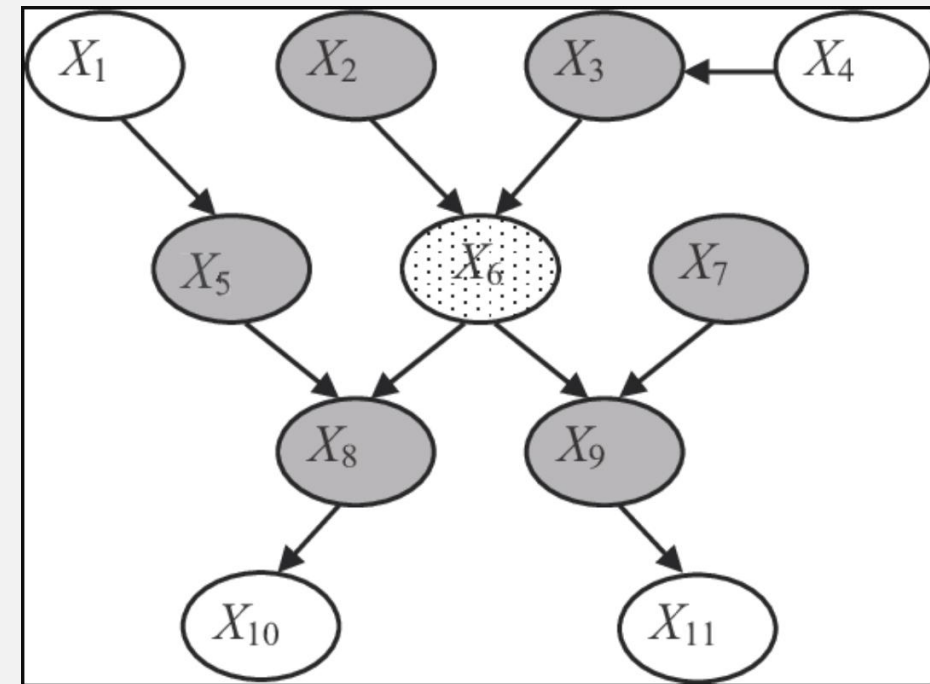
$$= \frac{\sum_{X_1, X_4, X_7, X_8, X_9, X_{10}, X_{11}} P(X_1, x_2, x_3, X_4, x_5, x_6, X_7, X_8, X_9, X_{10}, X_{11})}{\sum_{X_1, X_4, X_5, X_6, X_7, X_8, X_9, X_{10}, X_{11}} P(X_1, x_2, x_3, X_4, X_5, X_6, X_7, X_8, X_9, X_{10}, X_{11})}$$

$$= \frac{\sum_{X_1, X_4, X_7, X_8, X_9, X_{10}, X_{11}} P(X_1)P(x_5 \mid X_1)P(X_4)P(x_3 \mid X_4)P(X_7)P(x_2)P(x_6 \mid x_2, x_3)P(X_9 \mid x_6, X_7)P(X_8 \mid x_5, x_6)P(X_{10} \mid X_8)P(X_{11} \mid X_9)}{\sum_{X_1, X_4, X_5, X_6, X_7, X_8, X_9, X_{10}, X_{11}} P(X_1)P(X_5 \mid X_1)P(X_4)P(x_3 \mid X_4)P(X_7)P(x_2)P(X_6 \mid x_2, x_3)P(X_9 \mid X_6, X_7)P(X_8 \mid X_5, X_6)P(X_{10} \mid X_8)P(X_{11} \mid X_9)}$$

Some terms here sum to 1, and some terms cancel with denominator.

$$= P(x_6 \mid x_2, x_3) \frac{\sum_{X_1} P(X_1)P(x_5 \mid X_1)}{\sum_{X_1, X_5} P(X_1)P(X_5 \mid X_1)} \quad \text{Denominator sums to 1}$$

$$= P(x_6 \mid x_2, x_3)P(x_5) = P(x_6 \mid x_2, x_3)P(x_5 \mid x_2, x_3)$$



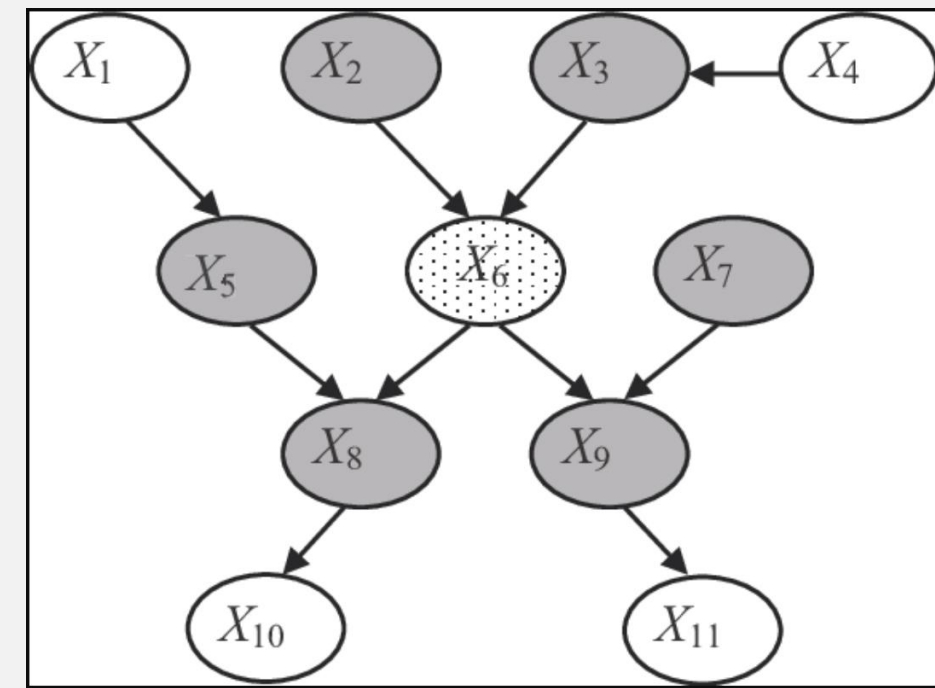
So, we could have just decoupled  $x_5$  and  $x_6$  from the beginning using the rule.

“a node is conditionally independent of all other nodes in the network, given its parents, children, and children’s parents—that is, given its **Markov blanket**”

What is  $P(x_6 \mid x_2, x_3, x_5, x_7, x_8, x_9)$  ? given the Markov blanket of  $X_6$

$$P(x_6 \mid x_2, x_3, x_5, x_7, x_8, x_9) = \frac{P(x_6, x_2, x_3, x_5, x_7, x_8, x_9)}{P(x_2, x_3, x_5, x_7, x_8, x_9)}$$

$$= \frac{\sum_{X_1, X_4, X_{10}, X_{11}} P(X_1, X_4, X_{10}, X_{11}, x_6, x_2, x_3, x_5, x_7, x_8, x_9)}{\sum_{X_1, X_4, X_6, X_{10}, X_{11}} P(X_1, X_4, X_6, X_{10}, X_{11}, x_2, x_3, x_5, x_7, x_8, x_9)}$$



$$= \frac{\sum_{X_1, X_4, X_{10}, X_{11}} P(X_1)P(x_5 \mid X_1)P(X_4)P(x_3 \mid X_4)P(x_7)P(x_2)P(x_6 \mid x_2, x_3)P(x_9 \mid x_6, x_7)P(x_8 \mid x_5, x_6)P(X_{10} \mid x_8)P(X_{11} \mid x_9)}{\sum_{X_1, X_4, X_6, X_{10}, X_{11}} P(X_1)P(x_5 \mid X_1)P(X_4)P(x_3 \mid X_4)P(x_7)P(x_2)P(X_6 \mid x_2, x_3)P(x_9 \mid X_6, x_7)P(x_8 \mid x_5, X_6)P(X_{10} \mid x_8)P(X_{11} \mid x_9)}$$

$$= \frac{P(x_6 \mid x_2, x_3)P(x_9 \mid x_6, x_7)P(x_8 \mid x_5, x_6)}{\sum_{X_6} P(X_6 \mid x_2, x_3)P(x_9 \mid X_6, x_7)P(x_8 \mid x_5, X_6)}$$

Notice that given the Markov blanket of  $X_6$ , we do not need to involve other variables to compute the query.

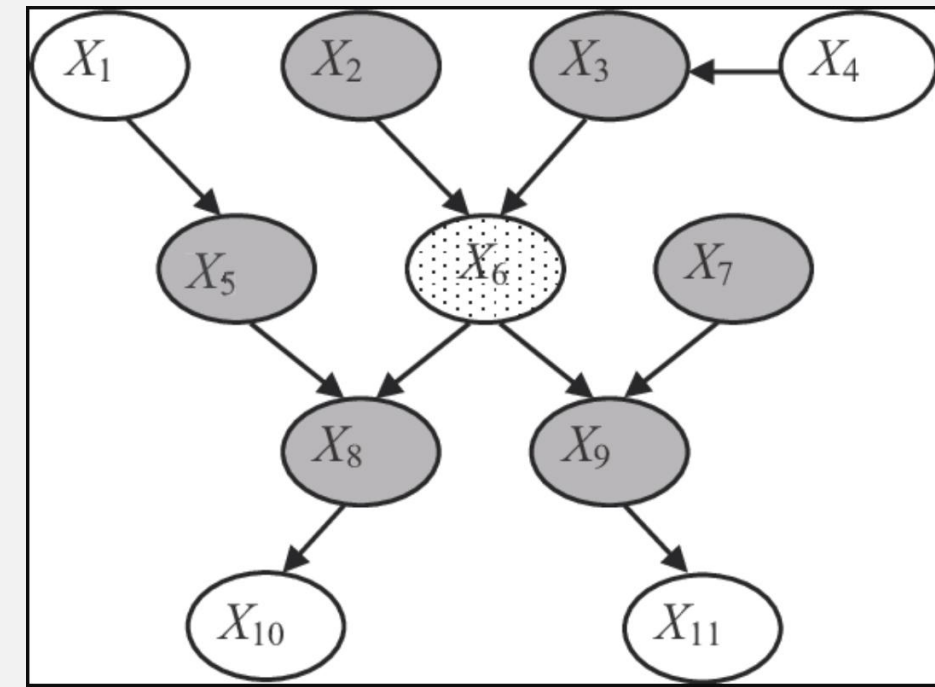
What is  $P(x_{11}, x_6 \mid x_2, x_3, x_5, x_7, x_8, x_9)$  ?

Using Markov blanket for  $x_6$  is given

$$= P(x_6 \mid x_2, x_3, x_5, x_7, x_8, x_9) P(x_{11} \mid x_2, x_3, x_5, x_7, x_8, x_9)$$

Using that parents of  $x_{11}$  are given

$$= P(x_6 \mid x_2, x_3, x_5, x_7, x_8, x_9) P(x_{11} \mid x_9)$$



- There is a more formal algorithm/process to determine which variables are independent given evidence in a Bayesian network called **d-separation**. Which can help a lot in reducing the computation work. See <http://web.mit.edu/jmn/www/6.034/d-separation.pdf> for a nice explanation and examples.
- **Main takeaway:** can use BN properties to decouple queries given evidence. This may reduce a lot of computation work.

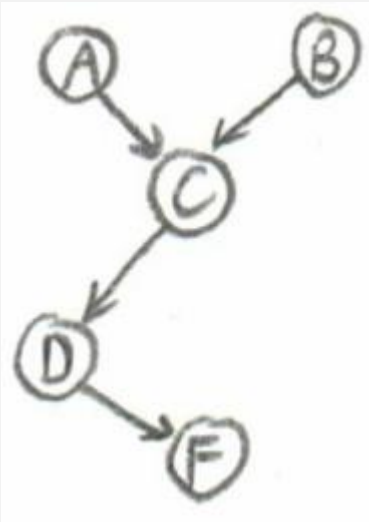
# D-Separation – technique to determine independence in a BN

- There is a more formal algorithm/process to determine which variables are independent given evidence in a Bayesian network called **d-separation**.
- Given a query  $P(Q \mid e)$ , D-separation works as follows:
  1. Draw the ancestral graph. Construct the “ancestral graph” of all variables mentioned in the probability expression ( $Q$  and  $e$ ). This is a reduced version of the original network, consisting only of the variables mentioned and all of their ancestors (parents, parents’ parents, etc.)
  2. Connect parents. For each pair of variables with a common child, draw an undirected edge (line) between them. (If a variable has more than two parents, draw lines between every pair of parents)
  3. Replace the directed edges (arrows) with undirected edges (lines).
  4. Delete the given variables  $e$  and their edges.
  5. Determine independence using available paths:
    - *Disconnected* variables in this graph, are guaranteed to be independent given the evidence.
    - *Connected* variables in this graph are **not** guaranteed to be independent given the evidence.
- See <http://web.mit.edu/jmn/www/6.034/d-separation.pdf> for a nice explanation and examples.

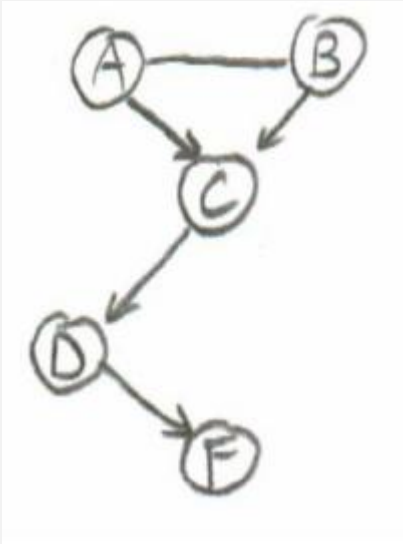
BN D-Separation Example [1]

Are A and B conditionally independent given D and F?  $A \perp\!\!\!\perp B \mid D, F$

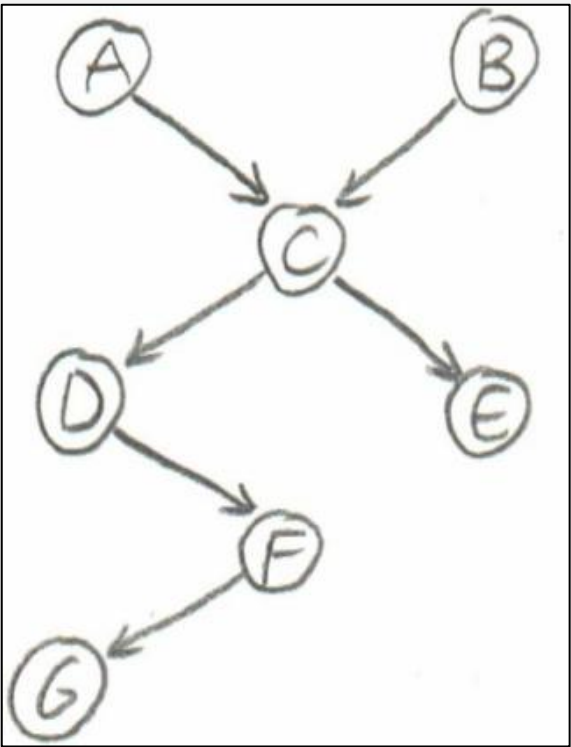
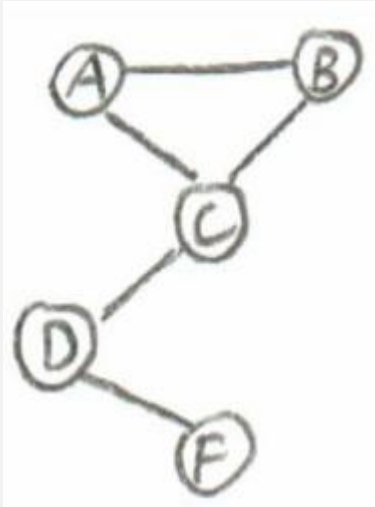
Step 1: ancestry graph



Step 2: connect parents

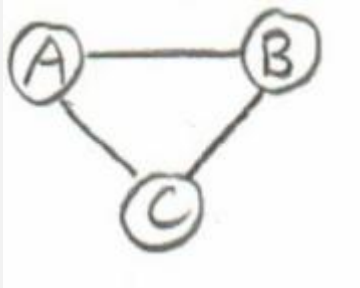


Step 3: make undirected



Bayesian network

Step 4: delete givens

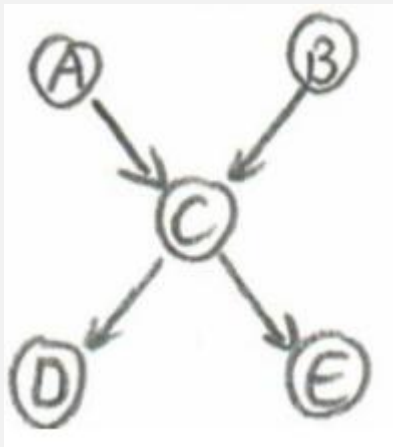


Answer: There is a path between A and B, so we cannot confirm the statement.

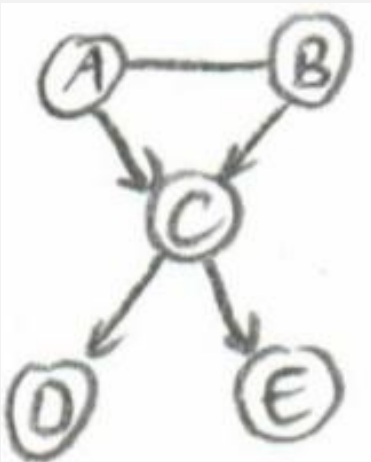
BN D-Separation Example [2]

Is  $D \perp\!\!\!\perp E \mid C$  ? Note that  $\perp\!\!\!\perp$  is independence symbol.

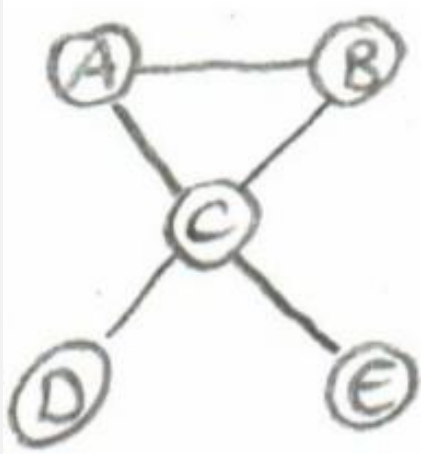
Step 1: ancestry graph



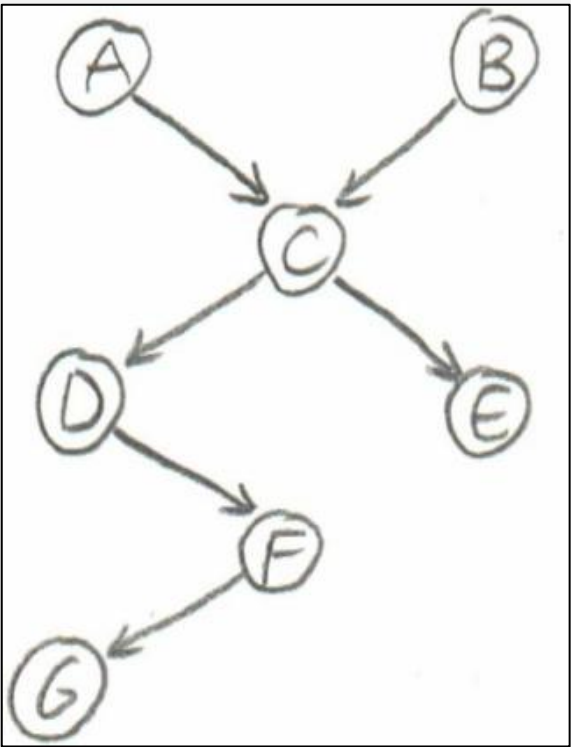
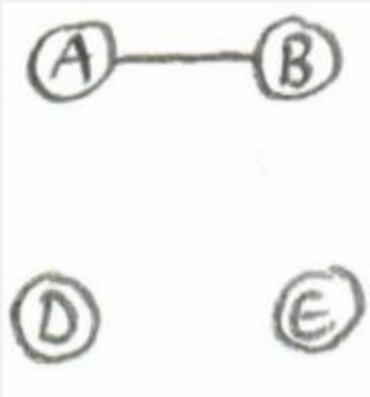
Step 2: connect parents



Step 3: make undirected



Step 4: delete givens



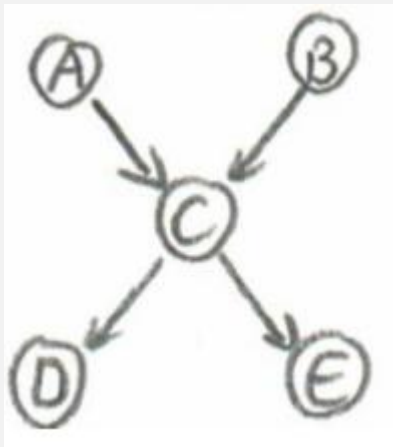
Bayesian network

Answer: There is no path between E and D, so we confirm the statement is true.

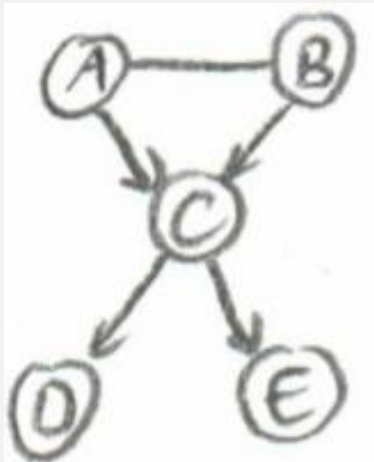
BN D-Separation Example [3]

Is  $D \perp\!\!\!\perp E$  ? Note no evidence given.

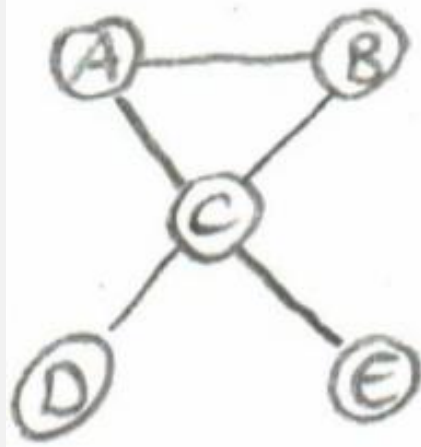
Step 1: ancestry graph



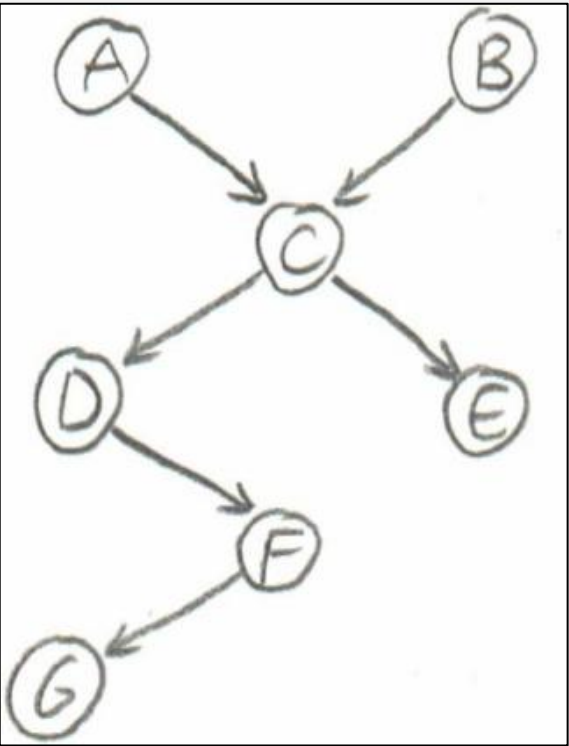
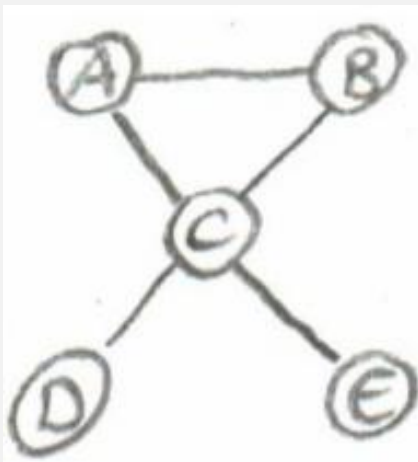
Step 2: connect parents



Step 3: make undirected



Step 4: delete givens, no givens so no deletions



Bayesian network

Answer: There is a path between E and D,  
so we cannot confirm the statement.

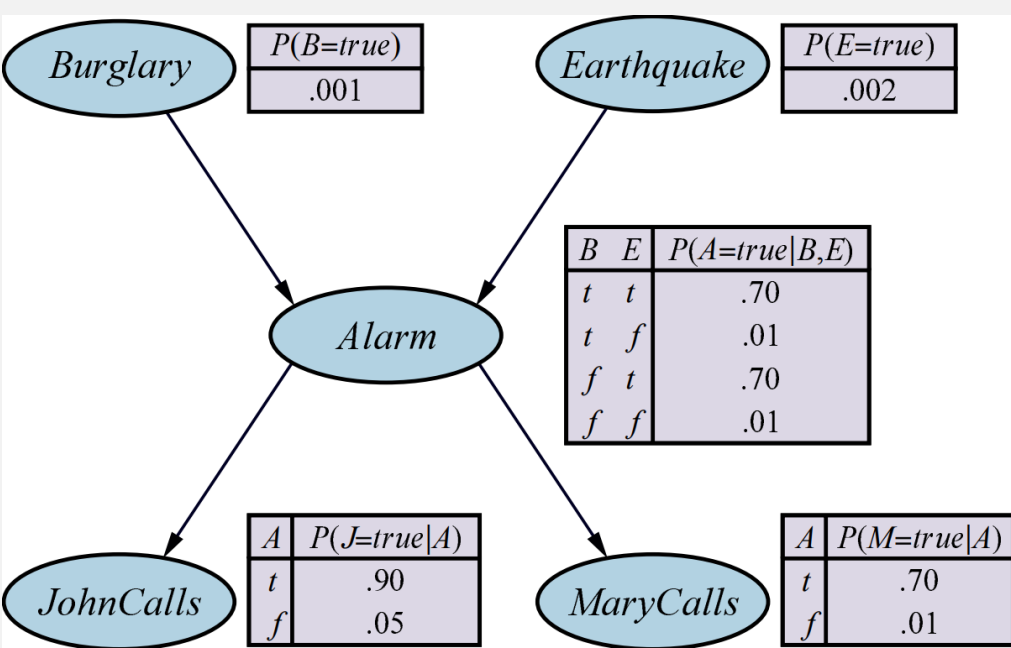
What is  $P(j)$  ?

$$P(j) = \sum_{B,E,A,M} P(B, E, A, j, M)$$

$$= \sum_{B,E,A,M} P(B)P(E)P(A \mid B, E)P(j \mid A)P(M \mid A)$$

$$= \sum_B P(B) \left[ \sum_E P(E) \left[ \sum_A P(A \mid B, E) P(j \mid A) \left[ \underbrace{\sum_M P(M \mid A)}_{\text{Sums to 1}} \right] \right] \right]$$

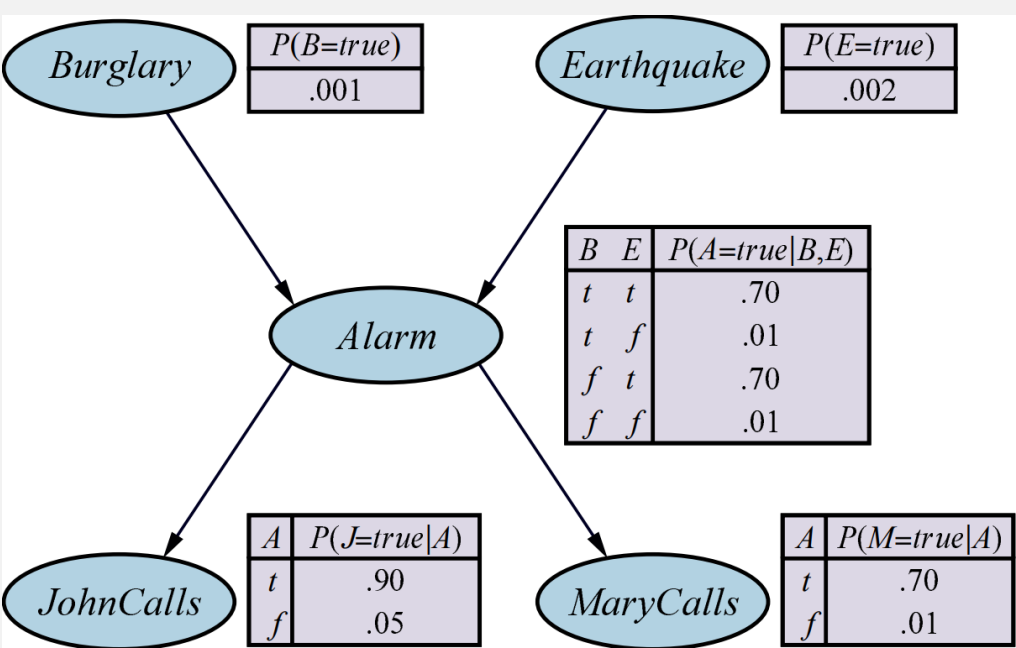
$$= \sum_B P(B) \left[ \sum_E P(E) \left[ \sum_A P(A \mid B, E) P(j \mid A) \right] \right]$$



With a BN, to compute  $P(X)$ , we just need to sum over the hidden variables that influence  $X$  **directly or indirectly**.  
J is influenced by A, B, E. In other words, what variables do I need to know in order to sample J.

What is  $P(j, b, e, m \mid a)$  ?

$$\begin{aligned} P(j, b, e, m \mid a) &= \frac{P(j, b, e, m, a)}{P(a)} \\ &= \frac{P(b)P(e)P(a \mid b, e)P(j \mid a)P(m \mid a)}{\sum_{B, E, J, M} P(B, E, a, J, M)} \\ &= \frac{P(b)P(e)P(a \mid b, e)P(j \mid a)P(m \mid a)}{\sum_{B, E, J, M} P(B)P(E)P(a \mid B, E)P(J \mid a)P(M \mid a)} \\ &= \frac{P(b)P(e)P(a \mid b, e)P(j \mid a)P(m \mid a)}{\sum_{B, E} P(B)P(E)P(a \mid B, E) \underbrace{\sum_J P(J \mid a)}_{\text{Sums to 1}} \underbrace{\sum_M P(M \mid a)}_{\text{Sums to 1}}} \\ &= \frac{P(b)P(e)P(a \mid b, e)P(j \mid a)P(m \mid a)}{\sum_{B, E} P(B)P(E)P(a \mid B, E)} \end{aligned}$$

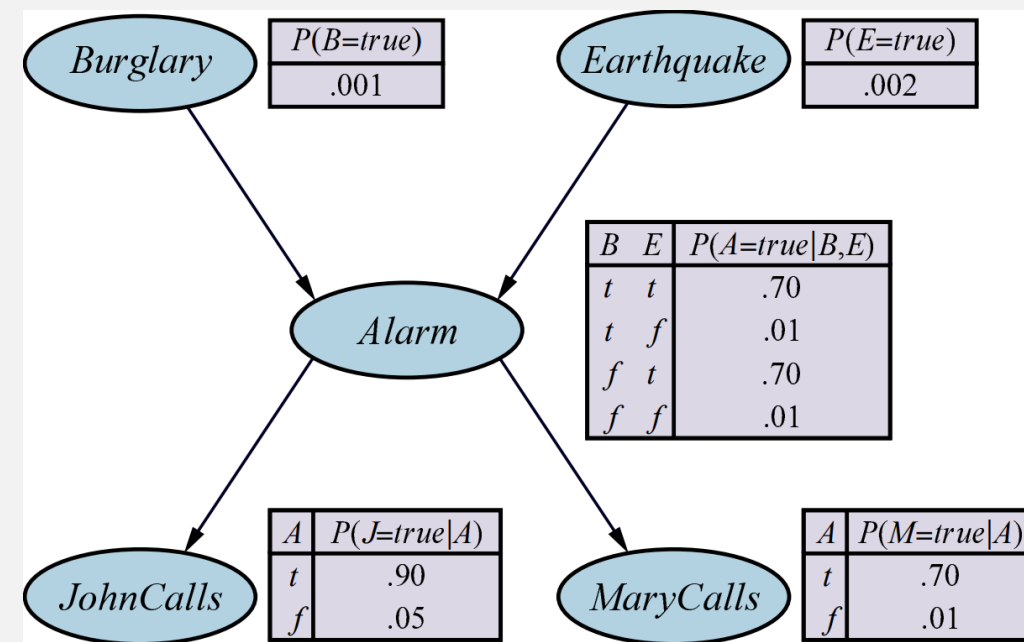


To compute  $P(a)$ , what variables do I need to know in order to sample  $A$ ? Just  $B$  and  $E$ .

What is  $P(\neg b, \neg m \mid a, \neg e)$  ?

$$P(\neg b, \neg m \mid a, \neg e) = \frac{P(\neg b, \neg m, a, \neg e)}{P(a, \neg e)}$$

$$= \frac{P(\neg b)P(\neg e)P(a \mid \neg b, \neg e)P(\neg m \mid a)}{\sum_B P(B)P(\neg e)P(a \mid B, \neg e)}$$

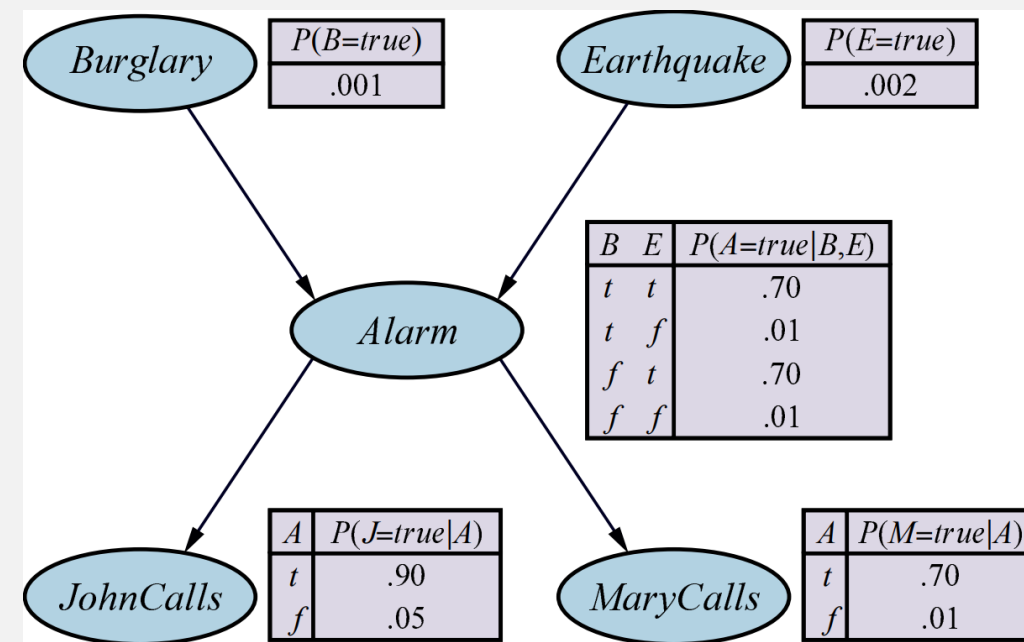


What is  $P(a \mid b, e, m, j)$  ?

$$P(a \mid b, e, m, j) = \frac{P(a, b, e, m, j)}{P(b, e, m, j)}$$

$$= \frac{P(b)P(e)P(a \mid b, e)P(m \mid a)P(j \mid a)}{\sum_A P(b)P(e)P(A \mid b, e)P(m \mid A)P(j \mid A)}$$

$$= \frac{P(a \mid b, e)P(m \mid a)P(j \mid a)}{\sum_A P(A \mid b, e)P(m \mid A)P(j \mid A)}$$



See this link for BN playground

[http://www.cs.man.ac.uk/~gbrown/bayes\\_nets/#](http://www.cs.man.ac.uk/~gbrown/bayes_nets/#)

# Constructing a Bayesian Network

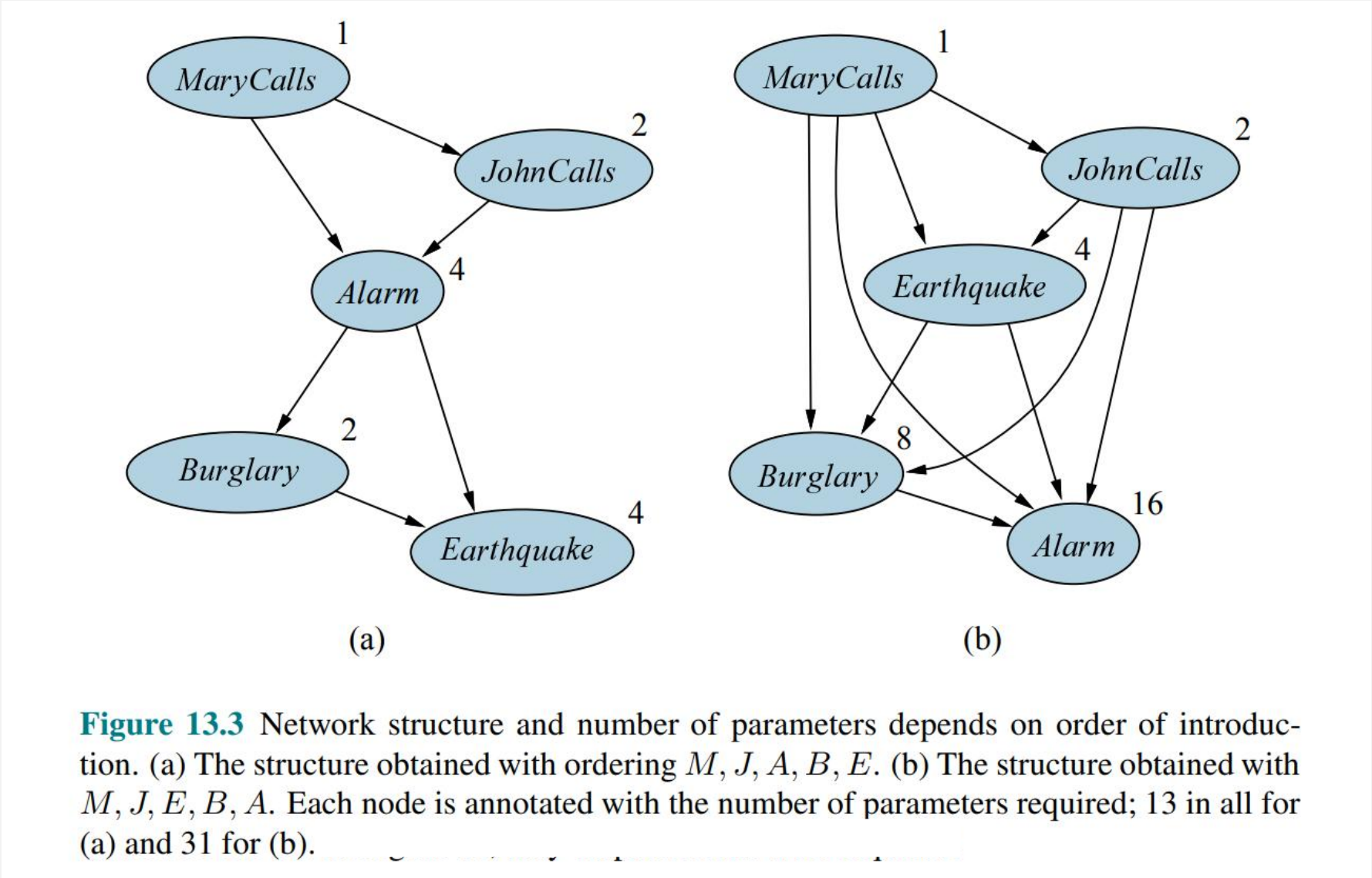
- How many graphs possible for  $n$  variables? How to find the **optimal** graph that best represents the process?
- One systematic way to construct a Bayesian network for  $n$  variables is as follows:
  1. Suppose you are given the full-joint table. Choose ordering of variables  $X_1, X_2, \dots, X_n$
  2. For  $i = 1$  to  $n$ 
    - add  $X_i$  to the network
    - Select minimal set of parents from  $X_1, X_2, \dots, X_{i-1}$  such that  $P(X_i | Parents(X_i)) = P(X_i | X_1, \dots, X_{i-1})$  is satisfied. How to do this in practice? Use gathered data samples to confirm they are approximately equal.
- This choice of parents guarantees:
$$P(X_1, X_2, \dots, X_n) = \prod_i P(X_i | X_1, X_2, \dots, X_{i-1}) = \prod_i P(X_i | Parents(X_i))$$
- However, note that the ordering you choose in step 1 will have an impact on the compactness of the resulting network. Typically, you want to order variables such that causes precede effects. This will give a more compact network.
- Finding **optimal** (most compact) Bayesian network is NP-hard (factorial and exponential terms).
- Domain knowledge can be used to construct the network directly. Or to give a better ordering of the variables that reflect causal relationships.
- In practice, we can use algorithms guided by heuristics and dataset to construct reasonably compact networks.

# Notes on constructing a Bayesian Network

- How many graphs possible for  $n$  variables? How to find the **optimal** graph that best represents the process?
- You can construct the network using an ordering on the variables such that causes precede effects. This will give a more compact network.
- However, in general, finding **optimal** (most compact) Bayesian network is NP-hard (factorial and exponential terms).
- Domain knowledge can be used to construct the network directly. Or to give a better ordering of the variables that reflect causal relationships.
- In practice, we can use algorithms guided by heuristics and dataset to construct reasonably compact networks.
- See the structure-learning algorithms at <https://pgmpy.org/>

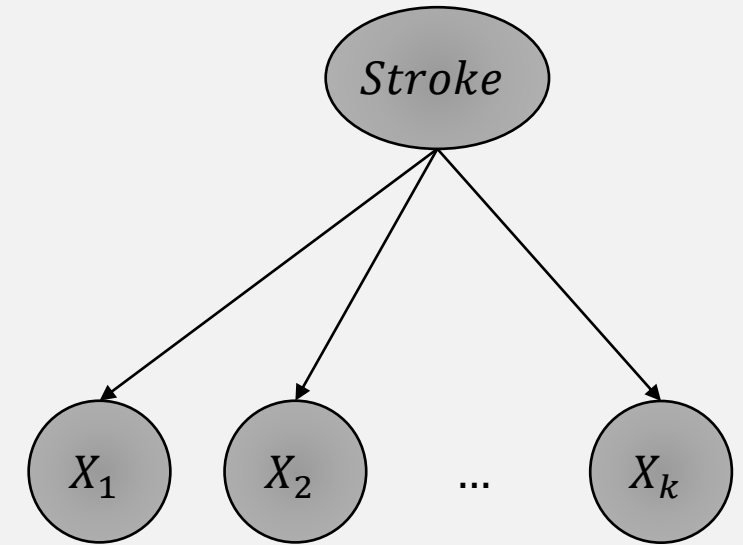
| Structure Learning |  |
|--------------------|--|
| PC with variants   |  |
| Hill-Climb Search  |  |
| Tree Search        |  |
| Max-Min Hill-Climb |  |
| Exhaustive Search  |  |

# Constructing a Bayesian Network – Impact of node ordering on size of network



## Using data to estimate BN parameters and structure.

- You are given a heart-stroke dataset.
- Let's use a naïve bayes BN to model heart-stroke.
- How many parameters does this BN have?
- We can use [pgmpy](#) to learn the BN parameters (probabilities) using a dataset.
- We can use algorithms guided by heuristics and dataset to construct reasonably compact networks. (won't cover these algorithms)
- How to compare between various BN structures for particular problem? Use a performance metric to compare between them. For example, we could have a held-aside heart-stroke set, and see which BN structure did the best in predicting stroke on this set.



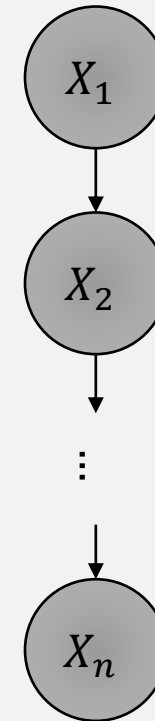
BN structure of naïve Bayes

# $n$ binary variables BN illustrations

- Suppose we have  $n$  binary variables.
- With no assumptions, we have the full-joint table with  $O(2^n)$  entries.
- If we assume all variables are independent of each other, this is a BN with no edges (all nodes have no connections).
- If we assume that  $X_i$  is influenced by  $X_{i-1}$ , we get a tree like BN.
- So, a BN structure is determined by the assumptions we make about the variable relationships. Typically, this is decided on by domain knowledge and intuition about the problem.



BN using all independence assumption.

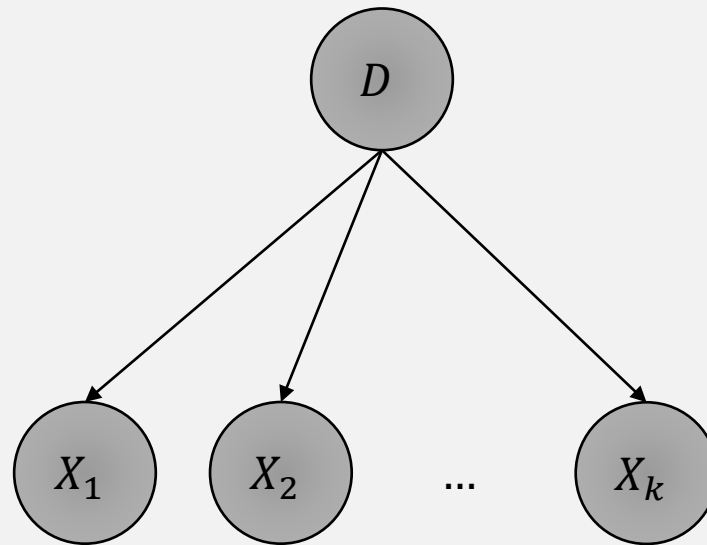


BN with assumption that  $X_i$  influenced by  $X_{i-1}$

# BN of Naïve Bayes

- Consider a naïve bayes where  $k$  different symptom variables  $X_1, X_2, \dots, X_k$  are conditionally independent of each other given  $D = \text{disease}$ .

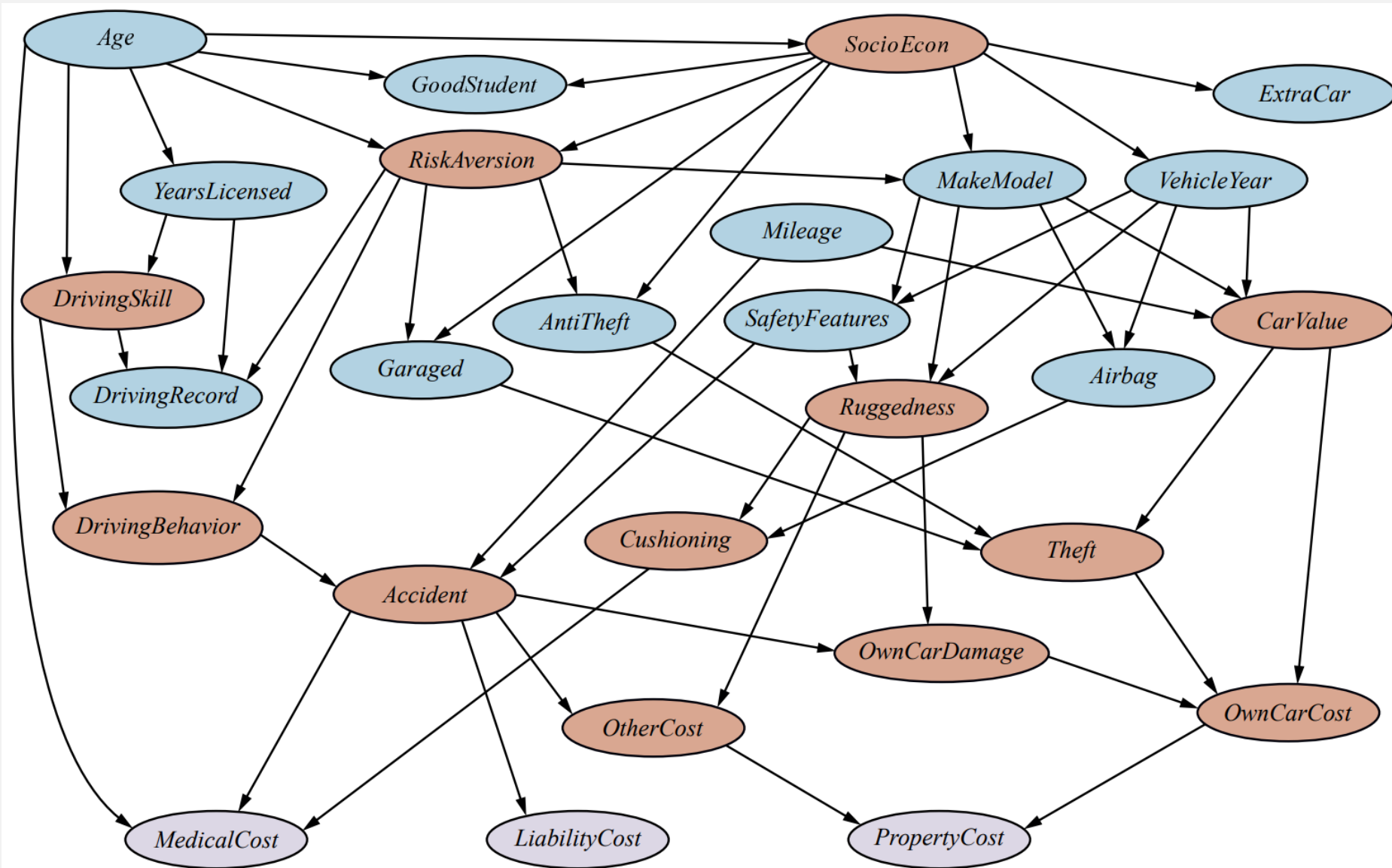
$$P(X_1, X_2, \dots, X_k \mid D) = \prod P(X_i \mid D)$$



BN structure of naïve Bayes

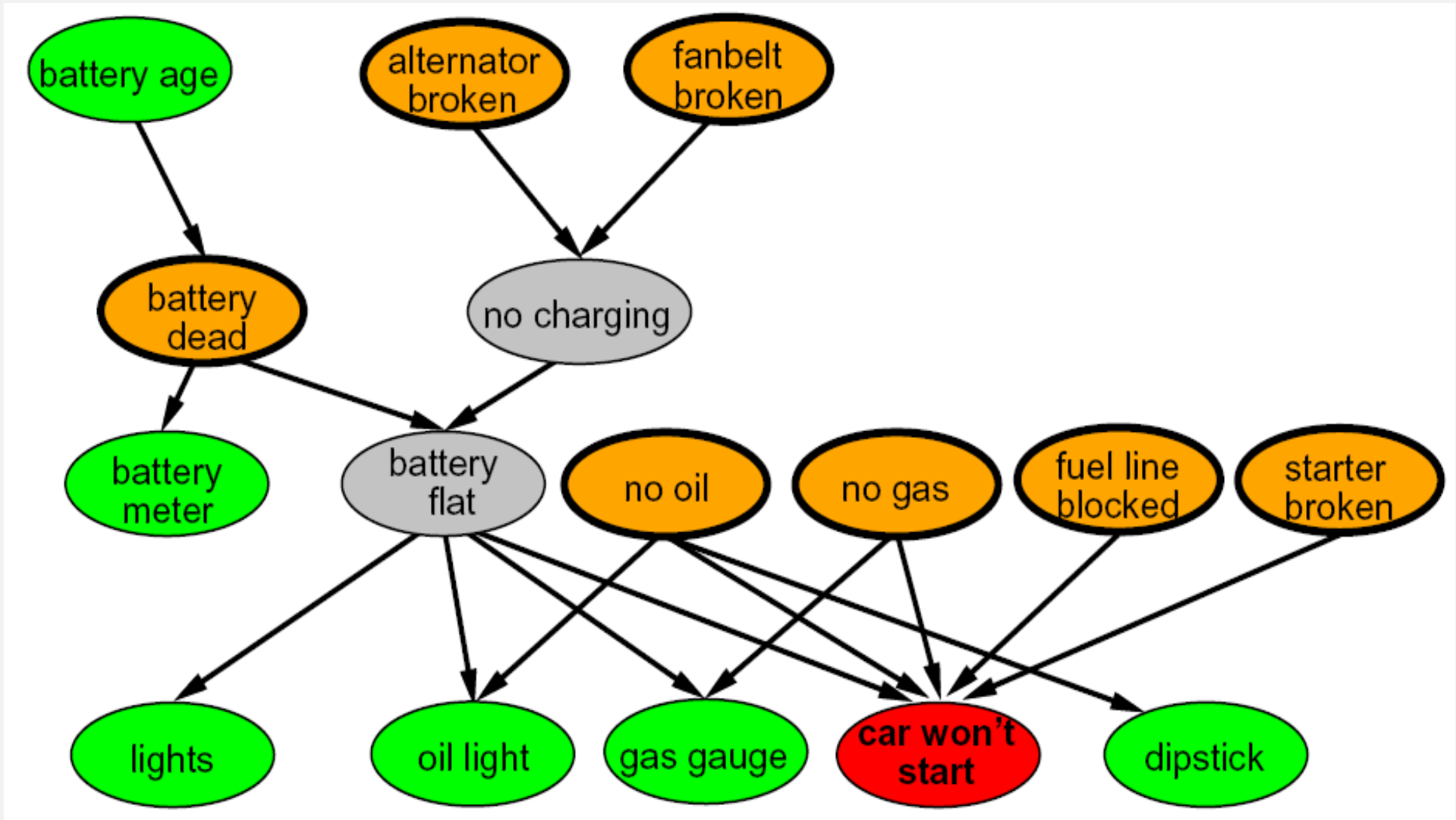
## Example BN: Car Insurance

- See [aima](#) textbook section 13.2.4.
- Gives example on how Bayes net is structured for a car insurance application.
- Can ask queries like what is the probability that medical cost will be high given information on the client.
- Typically in practice, the structure of a Bayesian network is determined by the designer using domain knowledge and intuition of what variables influence other variables.



**Figure 13.9** A Bayesian network for evaluating car insurance applications.

# BN Example – Car Won't Start



## Example Bayes' Net: Liver disorder diagnosis

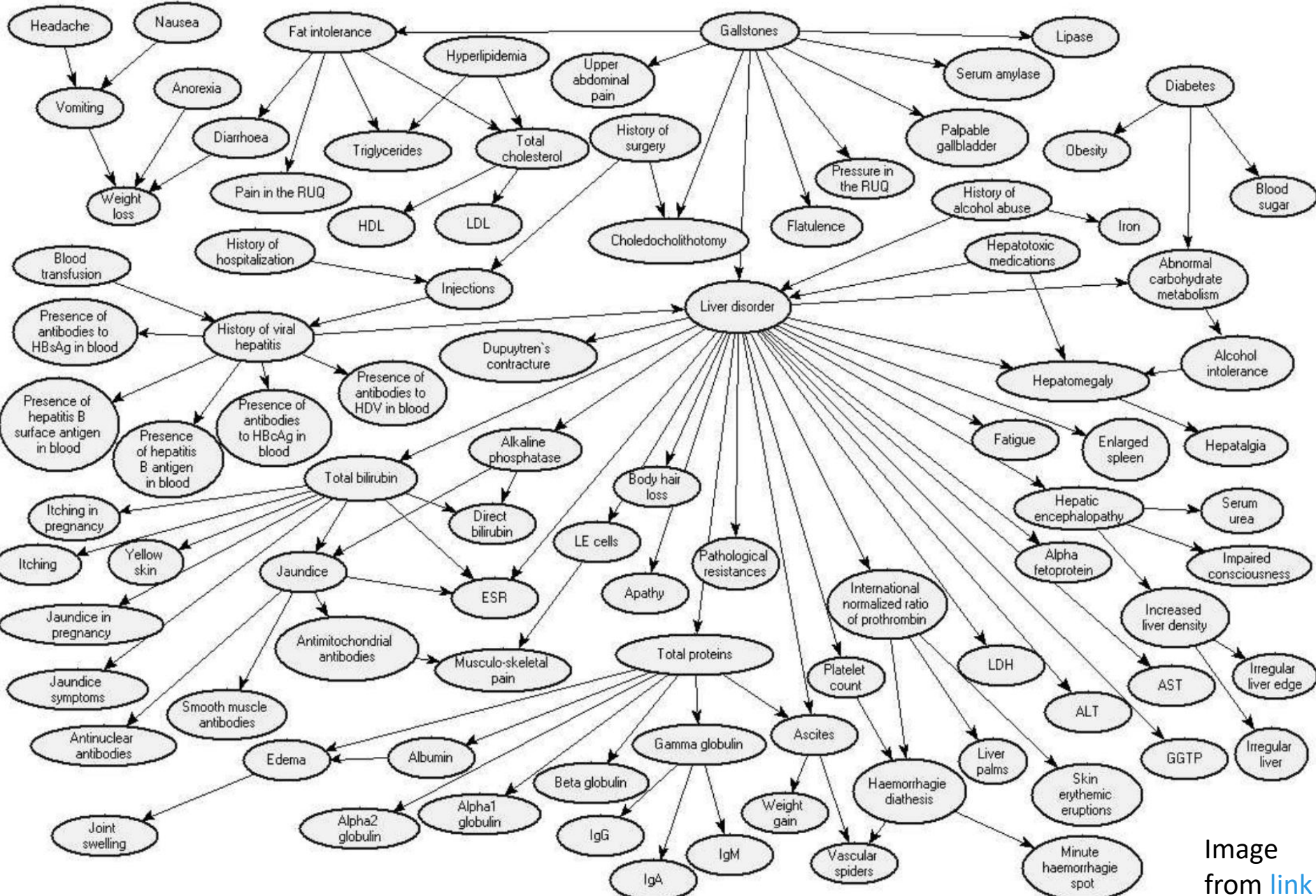


Image  
from [link](#)

Example Bayes' Net:  
Printer fault diagnosis

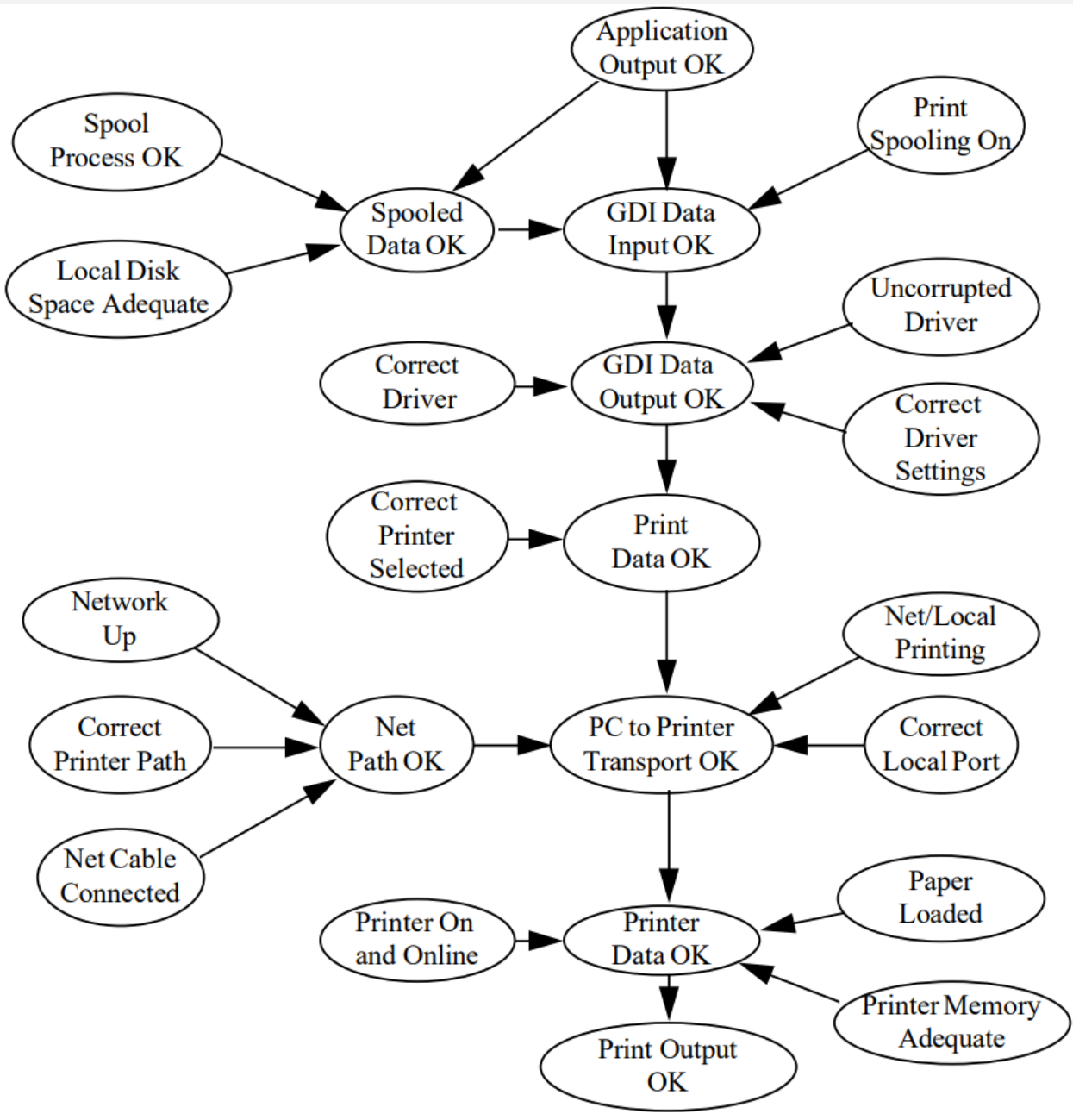


Image from [link](#)  
See also [link](#)

|                |                |     |     |
|----------------|----------------|-----|-----|
| 1,4            | 2,4            | 3,4 | 4,4 |
| 1,3            | 2,3            | 3,3 | 4,3 |
| 1,2<br>B<br>OK | 2,2            | 3,2 | 4,2 |
| 1,1<br>OK      | 2,1<br>B<br>OK | 3,1 | 4,1 |

- Let's consider a 4x4 grid environment with just breezes and pits.
- Let  $P_{i,j}$  denote pit in grid  $[i,j]$ . For syntax, we'll use  $p_{i,j}$  to denote there is a pit and  $\neg p_{i,j}$  for no pit in  $[i,j]$ .
- Let  $B_{i,j}$  denote breeze in grid  $[i,j]$ . For syntax, we'll use  $b_{i,j}$  to denote there is a breeze and  $\neg b_{i,j}$  for no breeze in  $[i,j]$ .
- In terms of probability modelling, the full-joint probability for this 4x4 grid environment involves 16 pit random variables + 16 breeze random variables = 32 random variables:

$$Prob(P_{1,1}, P_{1,2}, \dots, P_{4,4}, B_{1,1}, B_{1,2}, \dots, B_{4,4})$$

- We can reduce complexity by making the following reasonable assumptions:
  - $Prob(p_{i,j}) = 0.2$  this is saying that probability that a grid contains a pit is 0.2 in this environment.
  - A breeze in  $[i,j]$  is **caused** by some adjacent pit (can be caused by 1 pit, 2 pits, 3 pits, or 4 pits):

$$Prob(b_{i,j} \mid p_{i-1,j} \vee p_{i+1,j} \vee p_{i,j-1} \vee p_{i,j+1}) = 1 \quad \text{Note there are 7 possible ways for a breeze to be caused.}$$

- Special case: there can be no breeze in  $[i,j]$  iff there is no pit in adjacent grids (up, down, left, right):

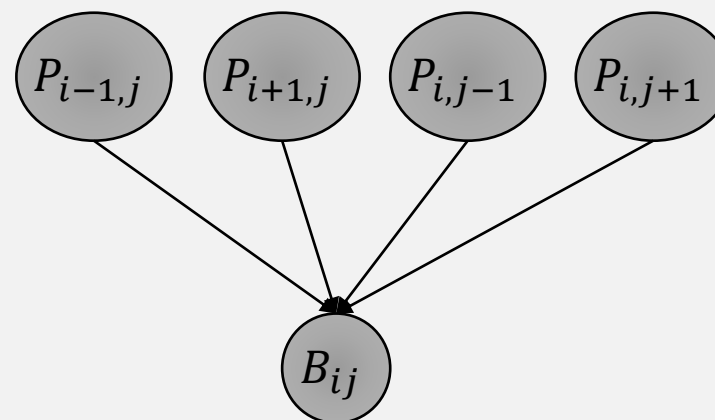
$$Prob(b_{i,j} \mid \neg p_{i-1,j}, \neg p_{i+1,j}, \neg p_{i,j-1}, \neg p_{i,j+1}) = 0$$

- Pits are independent of each other.
- We *cannot* say that breezes are independent of each other, because some breeze grids can share an adjacent pit. We *can* say that breezes are conditionally independent of each other *given* both their adjacent pit statuses.

- Using these assumptions, we get the following:

$$Prob(P_{11}, P_{12}, \dots, P_{44}, B_{11}, B_{12}, \dots, B_{4,4}) = \prod_{ij} Prob(P_{ij}) \prod_{ij} Prob(B_{ij} \mid P_{i-1,j}, P_{i+1,j}, P_{i,j-1}, P_{i,j+1})$$

- What does the BN look like for the 4x4 variant?



|                |                |     |     |
|----------------|----------------|-----|-----|
| 1,4            | 2,4            | 3,4 | 4,4 |
| 1,3            | 2,3            | 3,3 | 4,3 |
| 1,2<br>B<br>OK | 2,2            | 3,2 | 4,2 |
| 1,1<br>OK      | 2,1<br>B<br>OK | 3,1 | 4,1 |

- BN will have 16 nodes for the pits, and 16 nodes for the breezes. A breeze node will have at most 4 parent pits (2-3 parents for the boundary breezes). A pit node will have at most 4 breeze children (2-3 children for boundary pits).
- The full-joint has  $2^{32}$  versus the model with these assumptions has  $16 + 16 \times 2^4 = 272$  entries. For this particular model, we can get away with just 3 entries:
  - For  $Prob(P_{ij})$  we just need 1 entry since it is the same value for all pits 0.2
  - For  $Prob(B_{ij} \mid P_{i-1,j}, P_{i+1,j}, P_{i,j-1}, P_{i,j+1})$  we just need 2 entries since it is the same for all breezes (1 in case of pit in some adjacent, 0 in case of no adjacent pit).

Let us look at how to compute  $Prob(b_{i,j})$

$$Prob(b_{ij}) = \sum Prob(b_{ij}, P_{11}, \dots, P_{i-1,j}, P_{i+1,j}, P_{i,j-1}, P_{i,j+1}, \dots, P_{44}, B_{11}, \dots, B_{44}) \quad \text{Sum over hidden variables}$$

$$= \sum Prob(b_{ij}, P_{i-1,j}, P_{i+1,j}, P_{i,j-1}, P_{i,j+1} \mid \underbrace{P_{11}, \dots, P_{44}}_{\text{Here we mean the remaining model variables that are not } b_{ij}, P_{i-1,j}, P_{i+1,j}, P_{i,j-1}, P_{i,j+1}}, \underbrace{B_{11}, \dots, B_{44}}_{\text{Notice that these remaining variables do not influence } b_{ij}, P_{i-1,j}, P_{i+1,j}, P_{i,j-1}, P_{i,j+1}}) Prob(P_{11}, \dots, P_{44}, B_{11}, \dots, B_{44}) \quad \text{Product rule}$$

Here we mean the remaining model variables that are not  $b_{ij}, P_{i-1,j}, P_{i+1,j}, P_{i,j-1}, P_{i,j+1}$   
 Notice that these remaining variables do not influence  $b_{ij}, P_{i-1,j}, P_{i+1,j}, P_{i,j-1}, P_{i,j+1}$ .

$$= \sum Prob(b_{ij}, P_{i-1,j}, P_{i+1,j}, P_{i,j-1}, P_{i,j+1}) \sum Prob(P_{11}, \dots, P_{44}, B_{11}, \dots, B_{44}) \quad \text{We get independent summation terms. The second term are all hidden variables and summing this distribution will just give 1.}$$

$$= \sum Prob(b_{ij}, P_{i-1,j}, P_{i+1,j}, P_{i,j-1}, P_{i,j+1}) \quad \text{In other words, we just need to sum over hidden variables that influence the query and evidence variables.}$$

$$= \sum Prob(b_{ij} \mid P_{i-1,j}, P_{i+1,j}, P_{i,j-1}, P_{i,j+1}) Prob(P_{i-1,j}) Prob(P_{i+1,j}) Prob(P_{i,j-1}) Prob(P_{i,j+1})$$

Let us look at how to compute  $Prob(b_{i,j})$

$$Prob(b_{ij}) = \sum Prob(b_{ij}, P_{i-1,j}, P_{i+1,j}, P_{i,j-1}, P_{i,j+1})$$

In other words, we just need to sum over hidden variables that influence the query and evidence variables.

$$= \sum Prob(b_{ij} \mid P_{i-1,j}, P_{i+1,j}, P_{i,j-1}, P_{i,j+1}) Prob(P_{i-1,j}) Prob(P_{i+1,j}) Prob(P_{i,j-1}) Prob(P_{i,j+1})$$

We have  $2^4$  settings for hidden variables  $P_{i-1,j}, P_{i+1,j}, P_{i,j-1}, P_{i,j+1}$ .

$$= Prob(b_{ij} \mid p_{i-1,j}, p_{i+1,j}, p_{i,j-1}, p_{i,j+1}) Prob(p_{i-1,j}) Prob(p_{i+1,j}) Prob(p_{i,j-1}) Prob(p_{i,j+1}) + \dots \\ + Prob(b_{ij} \mid \neg p_{i-1,j}, \neg p_{i+1,j}, \neg p_{i,j-1}, \neg p_{i,j+1}) Prob(\neg p_{i-1,j}) Prob(\neg p_{i+1,j}) Prob(\neg p_{i,j-1}) Prob(\neg p_{i,j+1})$$

$$= (1 \times 0.2^4) + (1 \times 0.2 \times 0.2 \times 0.2 \times 0.8) + (1 \times 0.2 \times 0.2 \times 0.8 \times 0.2) + \dots + (0 \times 0.8^4)$$

The conditional prob. of breeze is 1 when some pit is present, otherwise zero.

$$= 0.5904$$

What about  $Prob(b_{ij} \mid \neg p_{i-1,j}, \neg p_{i,j+1})$

$$Prob(b_{ij} \mid \neg p_{i-1,j}, \neg p_{i,j+1}) = \sum Prob(b_{ij}, P_{i+1,j}, P_{i,j-1} \mid \neg p_{i-1,j}, \neg p_{i,j+1})$$

$$= \sum Prob(b_{ij} \mid \neg p_{i-1,j}, \neg p_{i,j+1}, P_{i+1,j}, P_{i,j-1}) Prob(P_{i+1,j}) Prob(P_{i,j-1})$$

We have  $2^2$  settings for hidden variables  $P_{i+1,j}, P_{i,j-1}$ .

$$= Prob(b_{ij} \mid \neg p_{i-1,j}, p_{i+1,j}, p_{i,j-1}, \neg p_{i,j+1}) Prob(p_{i+1,j}) Prob(p_{i,j-1}) + \dots \\ + Prob(b_{ij} \mid \neg p_{i-1,j}, \neg p_{i+1,j}, \neg p_{i,j-1}, \neg p_{i,j+1}) Prob(\neg p_{i+1,j}) Prob(\neg p_{i,j-1})$$

$$= (1 \times 0.2^2) + (1 \times 0.2 \times 0.8) + (1 \times 0.8 \times 0.2) + (0 \times 0.8^2)$$

$$= 0.36$$

We just need to sum over hidden variables that influence the query and evidence.  
Note that breezes are influenced (caused) by pits, but pits are not influenced by breezes.

What about  $Prob(b_{ij} \mid \neg p_{i-1,j}, \neg p_{i+1,j}, \neg p_{i,j+1})$

$$Prob(b_{ij} \mid \neg p_{i-1,j}, \neg p_{i+1,j}, \neg p_{i,j+1}) = \sum Prob(b_{ij} \mid \neg p_{i-1,j}, \neg p_{i+1,j}, \neg p_{i,j+1}, P_{i,j-1}) Prob(P_{i,j-1})$$

We have 2 settings for hidden variable  $P_{i,j-1}$ .

$$\begin{aligned} &= Prob(b_{ij} \mid \neg p_{i-1,j}, \neg p_{i+1,j}, p_{i,j-1}, \neg p_{i,j+1}) Prob(p_{i,j-1}) \\ &\quad + Prob(b_{ij} \mid \neg p_{i-1,j}, \neg p_{i+1,j}, \neg p_{i,j-1}, \neg p_{i,j+1}) Prob(\neg p_{i,j-1}) \end{aligned}$$

$$= (1 \times 0.2) + (0 \times 0.8)$$

$$= 0.2$$

As we gather more evidence of no adjacent pits, the probability of breeze goes down.

- Consider the situation in the figure. The agent explored grids [1,1], [1,2], and [2,1]. All were safe, breeze detected in [1,2] and [2,1].
- Using propositional logic as we saw in last topic, asking the KB, would just reply that there is a pit in [1,3] or [2,2] or [3,1], which is not helpful.
- Using a probability model, we can ask  $Prob(p_{13} \mid evidence)$ ,  $Prob(p_{22} \mid evidence)$ ,  $Prob(p_{31} \mid evidence)$ .
- Then move to grid that has lowest pit probability. This allows the agent to manage risk.
- What is the evidence in this current scenario? The info the agent knows about:
  - $\neg b_{11}$  no breeze in [1,1].
  - $\neg p_{11}$  no pit in [1,1].
  - $b_{12}$  breeze in [1,2].
  - $\neg p_{12}$  no pit in [1,2].
  - $b_{21}$  breeze in [2,1].
  - $\neg p_{21}$  no pit in [2,1].

|                |                |     |     |
|----------------|----------------|-----|-----|
| 1,4            | 2,4            | 3,4 | 4,4 |
| 1,3            | 2,3            | 3,3 | 4,3 |
| 1,2<br>B<br>OK | 2,2            | 3,2 | 4,2 |
| 1,1<br>OK      | 2,1<br>B<br>OK | 3,1 | 4,1 |

Using a probability model, what is  $Prob(p_{13} \mid evidence)$ ?

$$Prob(p_{13} \mid evidence) = Prob(p_{13} \mid \neg b_{11}, \neg p_{11}, b_{12}, \neg p_{12}, b_{21}, \neg p_{21})$$

$$= \frac{Prob(p_{13}, \neg b_{11}, \neg p_{11}, b_{12}, \neg p_{12}, b_{21}, \neg p_{21})}{Prob(\neg b_{11}, \neg p_{11}, b_{12}, \neg p_{12}, b_{21}, \neg p_{21})}$$

$$= \frac{\sum Prob(p_{13}, \neg p_{11}, \neg p_{12}, \neg p_{21}, P_{22}, P_{31}, \neg b_{11}, b_{12}, b_{21})}{\sum Prob(P_{13}, \neg p_{11}, \neg p_{12}, \neg p_{21}, P_{22}, P_{31}, \neg b_{11}, b_{12}, b_{21})}$$

We just need to sum over hidden variables that influence the query and evidence.  
Note that breezes are influenced (caused) by pits, but pits are not influenced by breezes.

$$= \frac{Prob(p_{13})Prob(\neg p_{11})Prob(\neg p_{12})Prob(\neg b_{11} \mid \neg p_{12}, \neg p_{21}) \sum Prob(P_{22})Prob(P_{31})Prob(b_{12} \mid \neg p_{11}, p_{13}, P_{22})Prob(b_{21} \mid \neg p_{11}, P_{22}, P_{31})}{Prob(\neg p_{11})Prob(\neg p_{12}) \sum Prob(P_{13})Prob(P_{22})Prob(P_{31})Prob(b_{12} \mid \neg p_{11}, P_{13}, P_{22})Prob(b_{21} \mid \neg p_{11}, P_{22}, P_{31})}$$

Common terms in numerator and denominator cancel out. Also,  $Prob(\neg b_{11} \mid \neg p_{12}, \neg p_{21}) = 1$

$$= \frac{Prob(p_{13}) \sum Prob(P_{22})Prob(P_{31})Prob(b_{12} \mid \neg p_{11}, p_{13}, P_{22})Prob(b_{21} \mid \neg p_{11}, P_{22}, P_{31})}{\sum Prob(P_{13})Prob(P_{22})Prob(P_{31})Prob(b_{12} \mid \neg p_{11}, P_{13}, P_{22})Prob(b_{21} \mid \neg p_{11}, P_{22}, P_{31})}$$

$$= \frac{Prob(p_{13}) \sum Prob(P_{22}) Prob(P_{31}) Prob(b_{12} \mid \neg p_{11}, p_{13}, P_{22}) Prob(b_{21} \mid \neg p_{11}, P_{22}, P_{31})}{\sum Prob(P_{13}) Prob(P_{22}) Prob(P_{31}) Prob(b_{12} \mid \neg p_{11}, P_{13}, P_{22}) Prob(b_{21} \mid \neg p_{11}, P_{22}, P_{31})}$$

$numerator = 0.2 \times ((0.2 \times 0.2) + (0.2 \times 0.8) + (0.8 \times 0.2)) = 0.072$

|                |                |     |     |
|----------------|----------------|-----|-----|
| 1,4            | 2,4            | 3,4 | 4,4 |
| 1,3            | 2,3            | 3,3 | 4,3 |
| 1,2<br>B<br>OK | 2,2            | 3,2 | 4,2 |
| 1,1<br>OK      | 2,1<br>B<br>OK | 3,1 | 4,1 |

The denominator are two terms, when  $P_{13} = true$  and  $P_{13} = false$ . We already calculated when  $P_{13} = true$ , it is the numerator.

$denominator = 0.072 + (0.8 \times ((0.2 \times 0.2) + (0.2 \times 0.8))) = 0.232$

$Prob(p_{13} \mid \neg b_{11}, \neg p_{11}, b_{12}, \neg p_{12}, b_{21}, \neg p_{21}) = \frac{0.072}{0.232} = 0.31$

By similar process we get:

$Prob(p_{31} \mid evidence) = 0.31$

$Prob(p_{22} \mid evidence) = 0.86$

Based on these probabilities, which grid should the agent explore first?

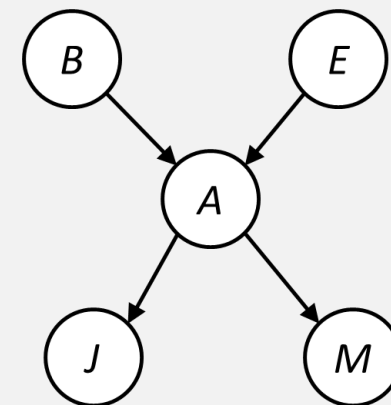
# Exact Inference by **enumeration** in Bayes Net

- Given the full joint distribution table  $P(X_1, \dots, X_n)$ , we can perform inference by enumeration by summing entries from the joint distribution:  $P(\mathbf{q}, \mathbf{e}) = \sum_{\mathbf{h}} P(\mathbf{q}, \mathbf{h}, \mathbf{e})$
- A Bayes network compactly represents the full-joint distribution using conditional probabilities. The space complexity is reduced to linear in number of variables  $O(n d^k)$ .
- What about time-complexity of answering a query in a BN?

- Burglar Network Example:

$$\begin{aligned} P(b, j, m) &= \sum_e \sum_a P(b, e, a, j, m) \\ &= \sum_e \sum_a P(b) P(e) P(a | b, e) P(j | a) P(m | a) \end{aligned}$$

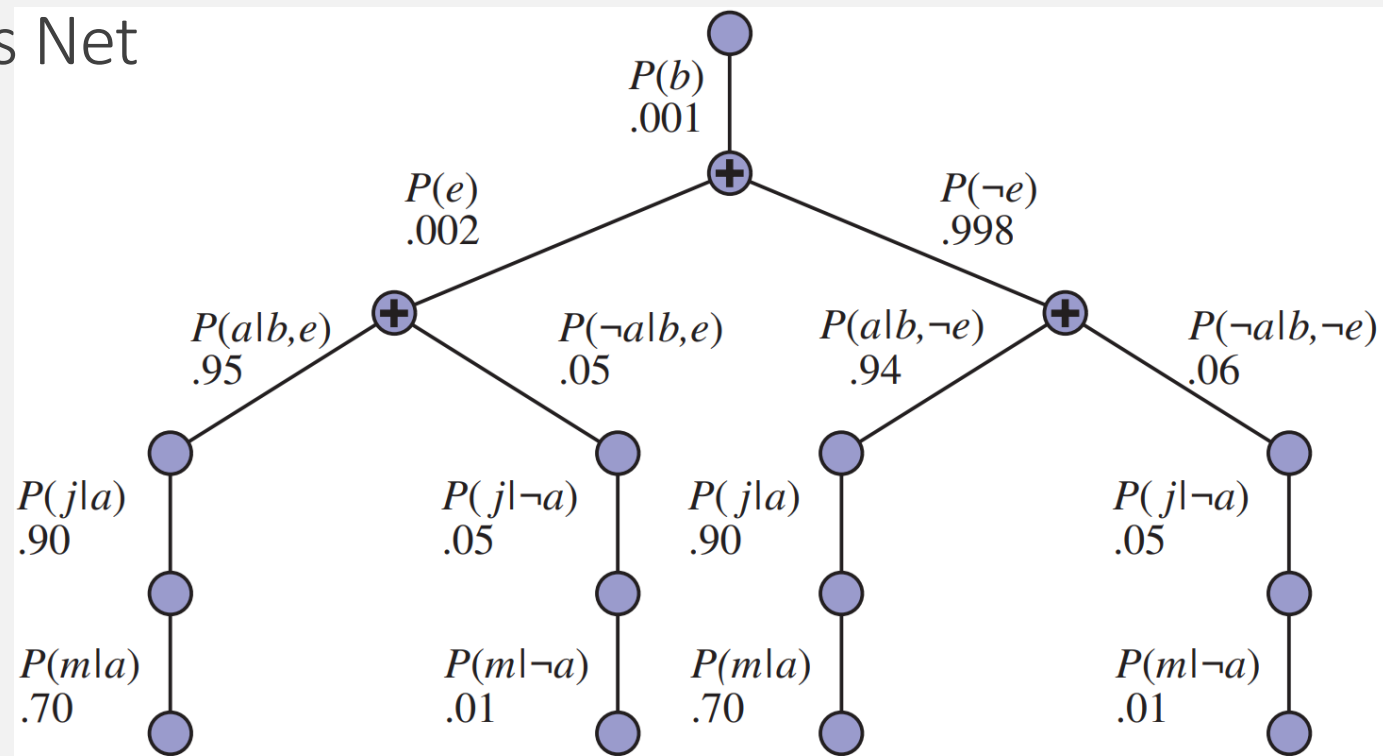
- Issues with this approach:
  - We have to *add* four terms (2 for e and 2 for a), each with 5 *multiplications*.
  - In worst-case, we have to sum over  $n - 1$  hidden variables; so  $O(2^n)$  *sums*, each a *product* of  $O(n)$  probabilities.
  - So, total worst-case runtime can be  $O(n 2^n)$ ; more work than full-joint due to products.



# Exact Inference by **enumeration** in Bayes Net

- So, total worst-case runtime can be  $O(n 2^n)$
- We can get rid of the  $n$  by re-arranging the terms and moving sums inward.

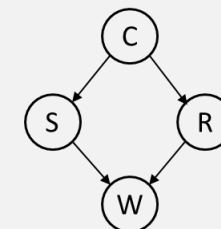
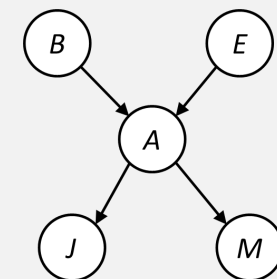
$$\begin{aligned} P(b \mid j, m) &= \sum_e \sum_a P(b, e, a, j, m) \\ &= \sum_e \sum_a P(b) P(e) P(a \mid b, e) P(j \mid a) P(m \mid a) \\ &= P(b) \sum_e P(e) \sum_a P(a \mid b, e) P(j \mid a) P(m \mid a) \end{aligned}$$



- Computation tree shown. This is a DFS approach:
  - Space complexity is linear in number of variables. (DFS)
  - For a general BN with  $n$  variables taking on  $d$  values and  $n - 1$  hidden variables query, worst-case time complexity is  $O(d^n)$ .
- Can we do better? If you look at the computation tree, you see that the product  $P(j \mid a)P(m \mid a)$  is computed twice. Similarly, for  $P(j \mid \neg a)P(m \mid \neg a)$ .
- The key to efficient Bayes net inference is to cache these and avoid re-computations. (hint: dynamic programming)

# Exact Inference in Bayes Net

- Can we do better? If you look at the computation tree, you see that the product  $P(j|a)P(m|a)$  is computed twice. Similarly, for  $P(j|\neg a)P(m|\neg a)$ .
- The key to efficient Bayes net inference is to avoid these re-computations. (hint: dynamic programming)
- **Variable elimination algorithm** does this
  - Dynamic programming-like approach; avoids re-computations.
  - We won't discuss details of variable elimination (for those interested see [video](#)).
  - Its time-complexity depends on structure of BN and order of variable processing.
  - *Worst-case* can still be exponential  $O(d^n)$ .
  - However, average-case it works well in practice with good ordering using heuristics.
- For the burglar network, we see there is **at most** one path between any two nodes.
  - Such networks are called **singly connected networks** or **polytrees**.
  - Time complexity of exact inference in polytrees is **linear** in the size of the network ( $O(n d^k)$ ;  $n$  variables and max  $k$  parents per variable). This can be done using variable elimination with a particular ordering that is efficient for polytrees.
- For **multiply connected networks** (which is the general case for BNs):
  - It is NP-hard. No known efficient algorithm exists. *Worst-case* is exponential  $O(d^n)$ .



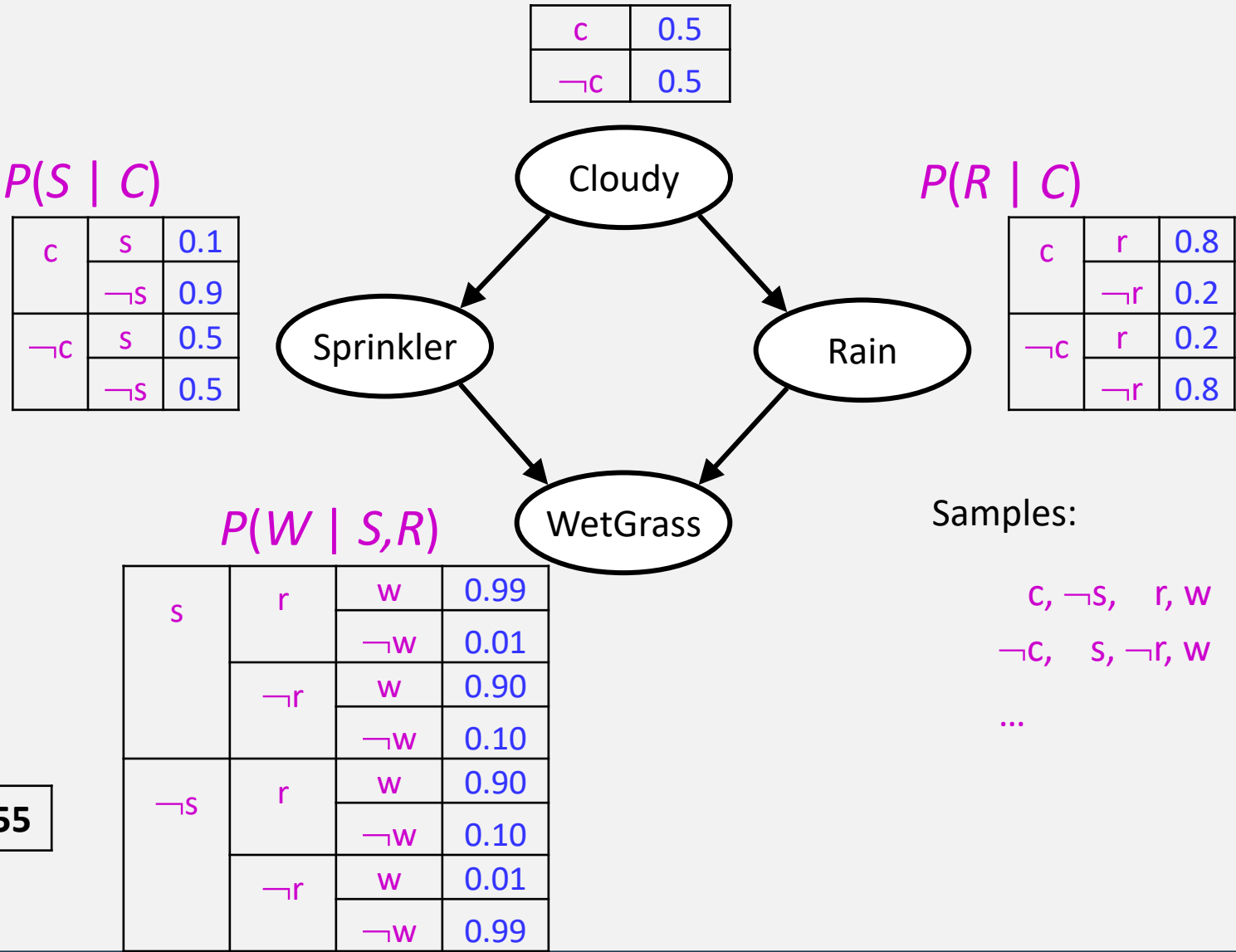
# Approximate Inference in BN

- Exact inference in BNs is intractable in general. However, we can consider approximate inference.
- Approximate inference in BN uses **random sampling** to compute approximate answers; the more samples we generate, the more accurate the answer.
- Random sampling algorithms are also called Monte-Carlo algorithms.
- Two main families of sampling algorithms:
  - Direct sampling algorithms like prior sampling, rejection sampling, and likelihood weighting.
  - Markov chain sampling algorithms like MCMC and Gibbs sampling. (won't cover this)
- We will discuss the three direct sampling algorithms to approximate probabilities. E.g. Given a BN network, what is the  $\text{Prob}(\text{QueryVars} \mid e)$ ?

# Approximate Inference in Bayes Net – Prior sampling

- In prior sampling, we sample from a BN based on its structure:
  - Sample variables with no parents first.
  - Then sample variables as their evidence becomes available.
- Consider the BN to the right. To generate a sample:
  - Sample in this order: Cloudy, Sprinkler, Rain, then WetGrass.
  - Another valid sample order: Cloudy, Rain, Sprinkler, then WetGrass.
- Suppose we sample the following values for  $u$ , what are the resulting samples from the BN? Use sample order: Cloudy, Sprinkler, Rain, then WetGrass.

|      |     |      |      |     |      |      |      |
|------|-----|------|------|-----|------|------|------|
| 0.35 | 0.7 | 0.15 | 0.85 | 0.9 | 0.25 | 0.42 | 0.55 |
|------|-----|------|------|-----|------|------|------|



# Approximate Inference in Bayes Net – Prior sampling

For  $i = 1, 2, \dots, n$  in topological order

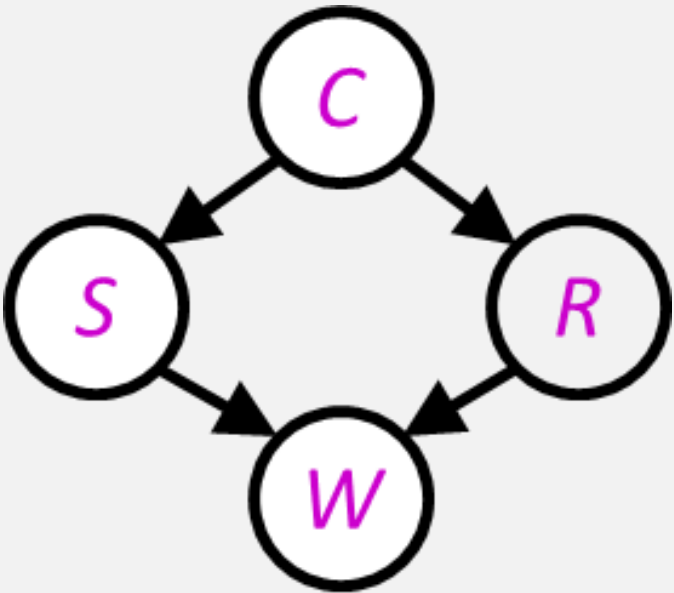
- Sample  $x_i$  from  $P(X_i \mid \text{Parents}(X_i))$

Return  $(x_1, x_2, \dots, x_n)$

# Approximate Inference in Bayes Net – Prior sampling Example [1]

- Consider BN to the right.
- Suppose we want to answer  $P(\neg s \mid r, w)$ ?
- We can use prior sampling. Suppose we sample the following:

|                             |   |                                                  |
|-----------------------------|---|--------------------------------------------------|
| $C, \neg S, r, w$           | ← | Among these samples,<br>how many have $\neg s$ ? |
| $C, s, r, w$                | ← |                                                  |
| $\neg C, s, r, \neg w$      |   |                                                  |
| $C, \neg S, r, w$           | ← |                                                  |
| $\neg C, \neg S, \neg r, w$ |   |                                                  |



- What is  $P(\neg s \mid r, w)$ ?
  - Grab samples consistent with evidence  $r, w$ . Among these samples, how many have  $\neg s$ ?
  - Thus,  $P(\neg s \mid r, w) \approx \frac{2}{3}$
  - This will get closer to the *exact* answer with more samples

# Approximate Inference in Bayes Net – Prior sampling Example

- Consider BN to the right.
- Suppose we want to answer  $P(w)$ ?
- We can use prior sampling. Suppose we sample the following:

$C, \neg S, r, w$

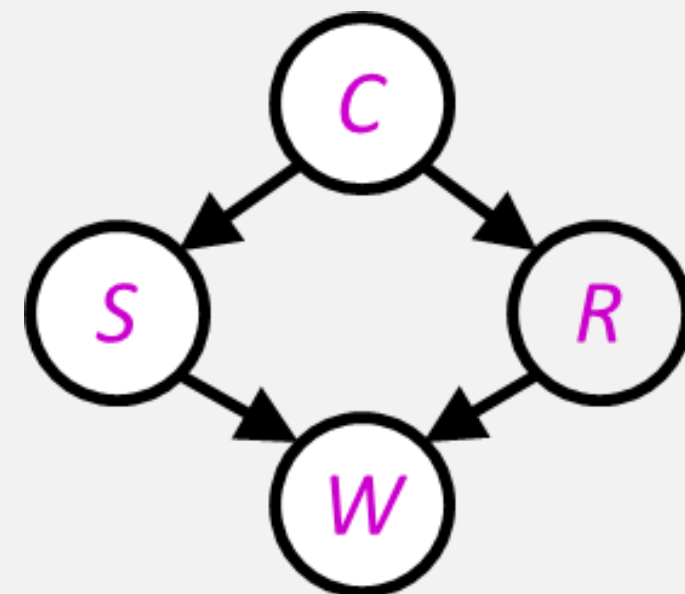
$C, s, r, w$

$\neg C, s, r, \neg w$

$C, \neg S, r, w$

$\neg C, \neg S, \neg r, w$

- What is  $P(w)$ ?
  - With *no evidence*, we consider all the samples. We have counts  $\langle w:4, \neg w:1 \rangle$
  - Thus,  $P(w) \approx \frac{4}{5}$
  - This will get closer to the *exact* answer with more samples



# Approximate Inference in Bayes Net – Prior sampling

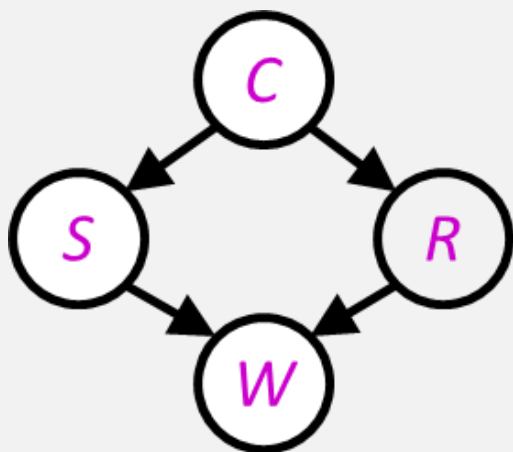
- Let  $S_{PS}(x_1, \dots, x_n)$  be the probability that prior-sampling generates the sample  $(x_1, \dots, x_n)$ . Then, prior sampling is sampling from the BN (which is equivalent to the joint distribution):

$$S_{PS}(x_1 \dots x_n) = \prod_{i=1}^n P(x_i | \text{Parents}(X_i)) = P(x_1 \dots x_n)$$

- Let  $N$  be the total number of samples generated.
- Let  $N_{PS}(x_1, \dots, x_n)$  be the number of times the sample  $(x_1, \dots, x_n)$  occurs.
- Then:  $P(x_1, \dots, x_n) \approx \frac{N_{PS}(x_1, \dots, x_n)}{N}$
- As we increase the number of samples generated, then we get:  $\lim_{N \rightarrow \infty} \frac{N_{PS}(x_1, \dots, x_n)}{N} = S_{PS}(x_1, \dots, x_n) = P(x_1, \dots, x_n)$
- So, the more samples we generate, the better this approximation becomes. It becomes exact in the limit; i.e. prior sampling is **consistent**.

# Approximate Inference in Bayes Net – Rejection Sampling

- **Rejection sampling** is *on-the-fly* application of prior sampling for estimating conditional probabilities
  - Let's say we want to compute  $P(c|\neg s, r)$
  - As we generate samples, samples with  $s$  or  $\neg r$  are not relevant (**reject/discard these samples**)
  - Prior sampling will generate  $N$  samples completely. Rejection sampling will discard inconsistent samples as it generates them.
  - So, rejection sampling can be used to generate  $N$  samples consistent with the evidence. Making use of all  $N$  samples for approximating the conditional query.
- Rejection sampling is **consistent**



$C, \neg S, r, W$   
 ~~$C, s$~~   
 $\neg C, \neg S, r, \neg W$   
 ~~$C, \neg S, \neg r$~~   
 $\neg C, \neg S, r, W$

As we generate a single sample, if we sample a value that is inconsistent with evidence, we can immediately stop and move on to next sample.

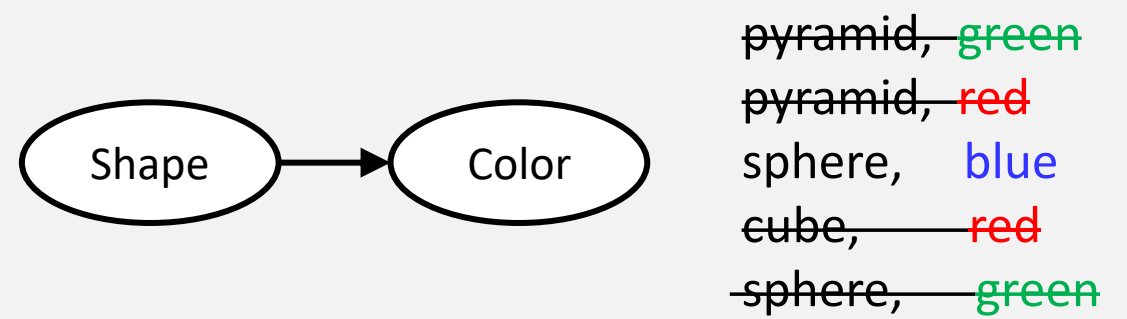
# Approximate Inference in Bayes Net – Rejection sampling

- Input: evidence  $e_1, \dots, e_k$
- For  $i=1, 2, \dots, n$ 
  - Sample  $X_i$  from  $P(X_i \mid \text{parents}(X_i))$
  - If  $x_i$  not consistent with evidence
    - Reject: Return, and no sample is generated in this cycle
- Return  $(x_1, x_2, \dots, x_n)$

$$P(X \mid e) \approx \frac{N_{PS}(X, e)}{N_{PS}(e)}$$

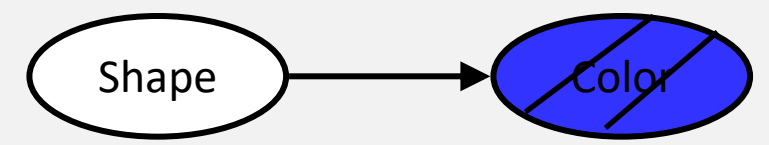
# Approximate Inference in Bayes Net – Likelihood Weighting

- Problem with rejection sampling:
  - If evidence is unlikely, rejects **lots** of samples
  - $P(\text{evidence})$  drops exponentially as number of evidence variables increases.
  - Thus, to get good approximation, it may require large number of samples particularly if evidence is rare.
  - For example, suppose that blue is a very rare color, then  $P(\text{Shape} | \text{Color}=\text{blue})$  would reject almost all samples.



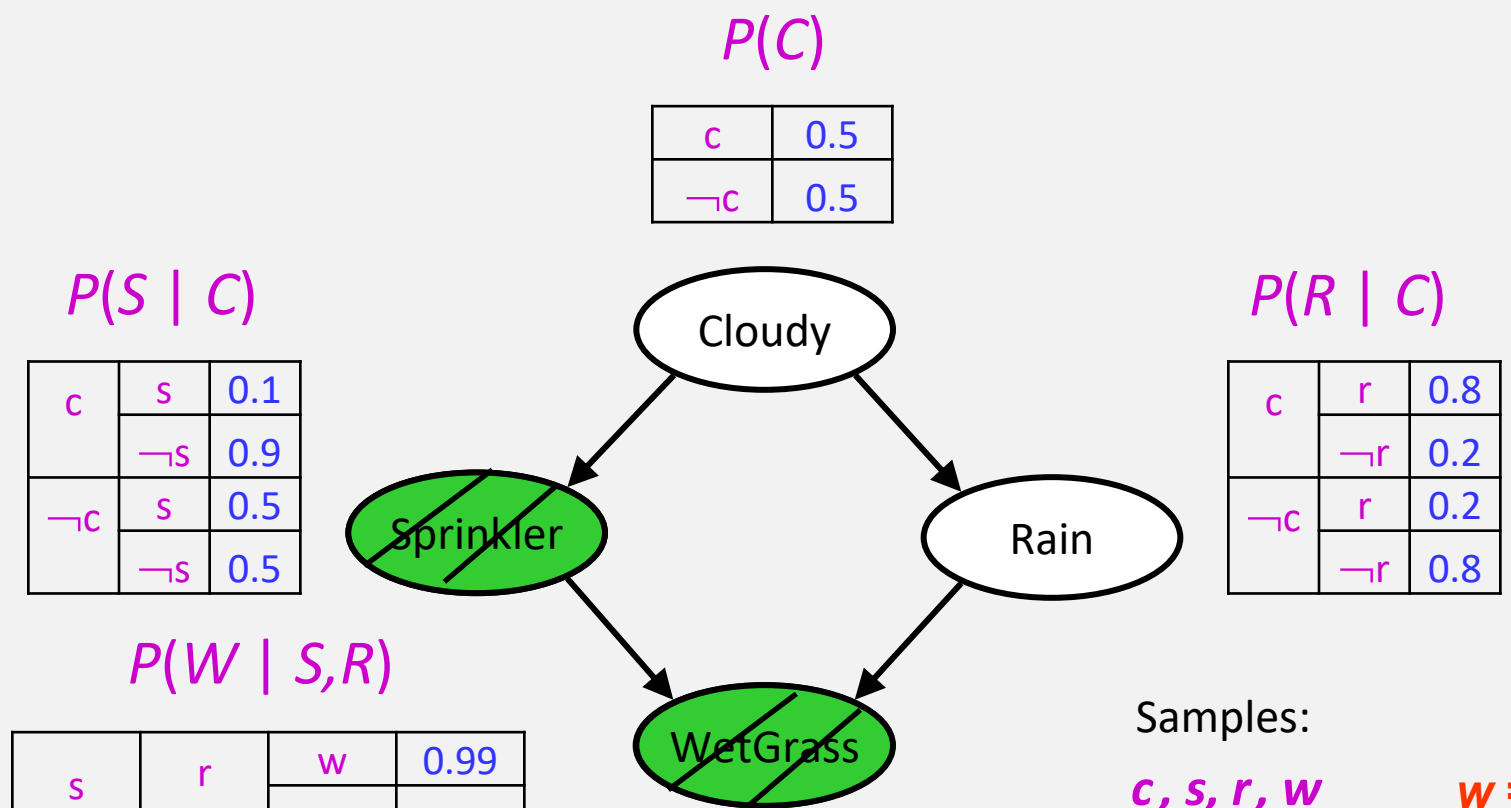
Idea: fix evidence variables, sample remaining variables

- Problem: sample distribution not consistent with true joint distribution! (we are not sampling directly from the BN)
- Solution: apply correction; **weight** each sample by probability of evidence variables given parents (using the BN probabilities)



|               |                                     |
|---------------|-------------------------------------|
| pyramid, blue | $w=P(\text{blue}   \text{pyramid})$ |
| pyramid, blue | $w=P(\text{blue}   \text{pyramid})$ |
| sphere, blue  | $w=P(\text{blue}   \text{sphere})$  |
| cube, blue    | $w=P(\text{blue}   \text{cube})$    |
| sphere, blue  | $w=P(\text{blue}   \text{sphere})$  |

# Approximate Inference in Bayes Net – Likelihood Weighting



- What is  $P(c, r | s, w)$ ?
- First fix  $s$  and  $w$  for each sample. Then sample remaining variables using BN.
- **Weight** each sample by probability of evidence variables given parents.
- To answer the query, use the weights of the samples.

Samples:

|                        |                                          |
|------------------------|------------------------------------------|
| $c, s, r, w$           | $w = 1.0 \times 0.1 \times 0.99 = 0.099$ |
| $\neg c, s, \neg r, w$ | $w = 1.0 \times 0.5 \times 0.9 = 0.45$   |
| $\neg c, s, r, w$      | $w = 1.0 \times 0.5 \times 0.99 = 0.495$ |

$$P(c, r | s, w) = \frac{\sum \text{weights of samples that have } (c, r, s, w)}{\sum \text{weights of samples that have } (s, w)}$$
$$= (0.099) / (0.099 + 0.45 + 0.495) = 0.0948$$

# Approximate Inference in Bayes Net – Likelihood Weighting

- Input: evidence  $e_1, \dots, e_k$
- $w = 1.0$
- for  $i=1, 2, \dots, n$ 
  - if  $X_i$  is an evidence variable
    - $x_i$  = observed value for  $X_i$
    - Set  $w = w * P(x_i \mid \text{parents}(X_i))$
  - else
    - Sample  $x_i$  from  $P(X_i \mid \text{parents}(X_i))$
- return  $(x_1, x_2, \dots, x_n), w$

- Suppose that  $P$  is the distribution given by the BN.
- We **fix** evidence  $\mathbf{e}$ , and sample remaining non-evidence  $\mathbf{z}$ . Thus, **sampling distribution** for a sample  $(\mathbf{z}, \mathbf{e})$ :

$$S_{WS}(\mathbf{z}, \mathbf{e}) = \prod_j P(z_j \mid \text{parents}(Z_j)) = N_{WS}(\mathbf{z}, \mathbf{e}) / N_{WS}(\mathbf{e})$$

- **Notice** this sampling distribution is not the same as the BN distribution! We are missing the evidence.
- To correct for this, we assign sample weights as:

$$w(\mathbf{z}, \mathbf{e}) = \prod_k P(e_k \mid \text{parents}(E_k))$$

- Putting together the sampling distribution and weights, then likelihood weighting is equivalent to the BN.

This is distribution of the BN

$$P(\mathbf{z}, \mathbf{e}) = \prod_j P(z_j \mid \text{parents}(Z_j)) \prod_k P(e_k \mid \text{parents}(E_k)) = S_{WS}(\mathbf{z}, \mathbf{e}) \cdot w(\mathbf{z}, \mathbf{e}) = \frac{N_{WS}(\mathbf{z}, \mathbf{e}) w(\mathbf{z}, \mathbf{e})}{N_{WS}(\mathbf{e})}$$

$$P(\mathbf{z} \mid \mathbf{e}) = \frac{P(\mathbf{z}, \mathbf{e})}{P(\mathbf{e})} = \frac{P(\mathbf{z}, \mathbf{e})}{\sum_X P(X, \mathbf{e})} = \frac{S_{WS}(\mathbf{z}, \mathbf{e}) \cdot w(\mathbf{z}, \mathbf{e})}{\sum_X S_{WS}(X, \mathbf{e}) w(X, \mathbf{e})} = \frac{N_{WS}(\mathbf{z}, \mathbf{e}) \cdot w(\mathbf{z}, \mathbf{e})}{\sum_X N_{WS}(X, \mathbf{e}) w(X, \mathbf{e})}$$

$$= \frac{\sum \text{weights of samples that have } (\mathbf{z}, \mathbf{e})}{\sum \text{weights of samples that have } (\mathbf{e})}$$

Note that  $X$  are all configurations of non-evidence variables.

# Notes on Approximate Inference using Sampling

- **Sampling-efficiency**: is the number of samples (or amount of work) an algorithm needs to approximate a value within some error tolerance.  $\text{abs}(\text{approxval} - \text{trueval}) \leq \epsilon$
- Sample-efficiency can be used to compare algorithms, we prefer an algorithm that does less work to approximate a value.
- Rejection sampling is more sample-efficient than prior sampling.
- Likelihood sampling uses all samples, so it is *more sample-efficient* than prior and rejection sampling. (prior and rejection discard some samples during generation)
- However, still degrades as the number of evidence variables increases, but less severe the rejection sampling. By degrades, we mean that to get within the specified tolerance, more samples are needed as the number of evidence variables increases.
- MCMC and Gibbs sampling alleviate degradation issues in likelihood sampling.

# Bayesian Network python package

- pgmpy python package for Bayesian networks.
- Supports Approximate (Sampling Based) and Exact inference

```
>>> from pgmpy.inference import VariableElimination
>>> from pgmpy.models import BayesianNetwork
>>> import numpy as np
>>> import pandas as pd
>>> values = pd.DataFrame(np.random.randint(low=0, high=2, size=(100
...                        columns=['A', 'B', 'C', 'D', 'E']))
>>> model = BayesianNetwork([('A', 'B'), ('C', 'B'), ('C', 'D'), ('B
>>> model.fit(values)
>>> inference = VariableElimination(model)
>>> phi_query = inference.query(['A', 'B'])
```

```
query(variables, evidence=None, virtual_evidence=None,
elimination_order='greedy', joint=True, show_progress=True)
```

```
rejection_sample(evidence=[], size=1, include_latents=False,
seed=None, show_progress=True, partial_samples=None) [so
```

```
likelihood_weighted_sample(evidence=[], size=1,
include_latents=False, seed=None, show_progress=True, n_jobs=-1)
```

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